I. SUPPLEMENTARY TABLES

TABLE S1

TABULAR REPRESENTATION OF THE 21 PLM/NLP-BASED EMBEDDINGS AND 19 CONVENTIONAL DESCRIPTORS ALONG WITH THEIR DIMENSIONS

Feature types	Features	Variant	Dimension
Conventional descriptors	AAC	-	20-D
	APAAC	-	26-D
	ASDC	-	400-D
	CKSAAGP-type-1	-	100-D
	CKSAAGP-type-2	-	100-D
	CKSAAP-type-1	-	1600-D
	CKSAAP-type-2	-	1600-D
	CTDC	-	39-D
	CTDD	-	195-D
	CTDT	-	39-D
	CTriad	-	343-D
	DDE	-	400-D
	DPC-type-1	-	400-D
	DPC-type-2	-	400-D
	DistancePair	-	20-D
	GDPC-type-2	-	25-D
	PAAC	-	23-D
	PDE	-	48-D
	QSOrder	-	46-D
PLM/NLP-based embeddings	Bepler	-	121-D
	CPCProt	-	512-D
	ESM1b	esm1b_t33_650M_UR50S	1280-D
	ESM1v	esm1v_t33_650M_UR90S	1280-D
	ESM2	esm2-t12-35M_UR50D	480-D
		esm2-t30-150M_UR50D	640-D
		esm2-t33-650M_UR50D	1280-D
		esm2-t36-3B_UR50D	2560-D
		esm2-t6-8M_UR50D	320-D
	ESM1	esm1_t34_670M_UR50S	1280-D
	FastText	-	512-D
	GloVe	-	512-D
	PRNN	-	1024-D
	PTAB	-	4096-D
	PTBB	-	1024-D
	PTB	-	1024-D
	PTU	-	1024-D
	PTXU	-	1024-D
	PTXLU	-	1024-D
	Seq2Vec	-	1024-D
	Word2Vec	-	512-D

TABLE S2

PARAMETER SEARCH RANGE FOR TRAINING HLPPRED 2.0 FRAMEWORK

Network	Hyperparameter	Search Range (Classification and Regression)
Trainer	Epochs	100
	Learning Rate	0.0001, 0.001
	Batch Size	8, 16
Transformer Encoder (TE)	Number of Multi-head SA (h_{enc})	4,8
	Transformer layers (L_{enc})	4, 6,8
	d_{target}	128, 256, 512
Multi-scale convolution localization (MSCL)	Kernel Size (k_i)	[1,3,5]
	Dilation (d)	1
	Stride (s)	1
Classifier / Regressor	Normalization	BatchNorm, LayerNorm
	Dropout Layer	0.1, 0.3
	FC Layer Channels	128,64,32

TABLE S3

PERFORMANCE COMPARISON OF HLPPRED 2.0 WITH EXISTING HEMOLYTIC PEPTIDE PREDICTORS ON THE COMPLETE INDEPENDENT DATASET (N=386)

Method	Model	N	AUC	MCC	ACC	Sn	Sp	Precision	F1-score	P-value (MCC)
HLPpred 2.0	**Hybrid DL classifier**	**386**	**0.9546** *(0.9364–0.9723)*	**0.7654** *(0.7028–0.8282)*	**0.8834** *(0.8523–0.9145)*	**0.8659** *(0.8156–0.9162)*	**0.8986** *(0.8551–0.9372)*	**0.8807** *(0.8387–0.9235)*	**0.8732** *(0.8371–0.9071)*	—
HemoPI2 (With MERCI)	ESM2-t6	386	0.9195 *(0.8918–0.9441)*	0.6515 *(0.5729–0.7289)*	0.8264 *(0.7876–0.8653)*	0.8212 *(0.7654–0.8771)*	0.8309 *(0.7778–0.8792)*	0.8077 *(0.7604–0.8571)*	0.8144 *(0.7718–0.8547)*	0.0040
HemoPI2 (With MERCI)	RF	386	0.9201 *(0.8934–0.9449)*	0.6732 *(0.5991–0.7459)*	0.8368 *(0.8005–0.8731)*	0.8436 *(0.7877–0.8939)*	0.8309 *(0.7778–0.8792)*	0.8118 *(0.7668–0.8611)*	0.8274 *(0.7875–0.8658)*	0.0142
HemoPI2 (Without MERCI)	ESM2-t6	386	0.8747 *(0.8377–0.9091)*	0.5895 *(0.5087–0.6702)*	0.7953 *(0.7539–0.8342)*	0.7933 *(0.7318–0.8492)*	0.7971 *(0.7391–0.8502)*	0.7717 *(0.7243–0.8232)*	0.7824 *(0.7380–0.8248)*	<.001
HemoPI2 (Without MERCI)	RF	386	0.8882 *(0.8546–0.9198)*	0.6168 *(0.5380–0.6945)*	0.8083 *(0.7694–0.8472)*	0.8212 *(0.7654–0.8771)*	0.7971 *(0.7440–0.8502)*	0.7778 *(0.7320–0.8280)*	0.7989 *(0.7569–0.8392)*	<.001
HLPpred-Fuse	ERT-based meta model	386	0.5740 *(0.5206–0.6274)*	0.0064 *(-0.0930–0.1063)*	0.4689 *(0.4482–0.4922)*	0.9497 *(0.9162–0.9777)*	0.0531 *(0.0242–0.0870)*	0.4645 *(0.4528–0.4762)*	0.6239 *(0.6073–0.6396)*	<.001
HemoPI	SVM	386	0.6349 *(0.5776–0.6891)*	0.2327 *(0.1379–0.3289)*	0.6192 *(0.5777–0.6606)*	0.3687 *(0.2961–0.4413)*	0.8357 *(0.7826–0.8841)*	0.6600 *(0.5800–0.7416)*	0.4731 *(0.4000–0.5431)*	<.001
HemoPred	Default	386	0.6923 *(0.6399–0.7441)*	0.1983 *(0.1037–0.2918)*	0.5648 *(0.5233–0.6062)*	0.8547 *(0.7989–0.9050)*	0.3140 *(0.2512–0.3816)*	0.5186 *(0.4916–0.5470)*	0.6456 *(0.6134–0.6768)*	<.001
HemoNet	CNN-based	386	0.6532 *(0.5990–0.7070)*	0.2484 *(0.1524–0.3424)*	0.6140 *(0.5674–0.6606)*	0.7318 *(0.6648–0.7989)*	0.5121 *(0.4444–0.5797)*	0.5647 *(0.5250–0.6059)*	0.6375 *(0.5917–0.6809)*	<.001
Transformer-CNN	Single Architecture	386	0.5852 *(0.5268–0.6405)*	0.1442 *(0.0454–0.2377)*	0.5415 *(0.4974–0.5829)*	0.8324 *(0.7765–0.8827)*	0.2899 *(0.2271–0.3527)*	0.5034 *(0.4759–0.5308)*	0.6274 *(0.5929–0.6593)*	<.001
Transformer-CNN	Combined Architecture	386	0.5673 *(0.5110–0.6246)*	0.0855 *(-0.0107–0.1787)*	0.5000 *(0.4689–0.5311)*	0.9106 *(0.8659–0.9497)*	0.1449 *(0.1013–0.1932)*	0.4794 *(0.4618–0.4971)*	0.6281 *(0.6047–0.6502)*	<.001
PeptideBERT	Default	386	0.8167 *(0.7728–0.8587)*	0.4217 *(0.3419–0.4986)*	0.6917 *(0.6554–0.7280)*	0.3911 *(0.3184–0.4637)*	0.9517 *(0.9227–0.9807)*	0.8750 *(0.8052–0.9412)*	0.5405 *(0.4653–0.6109)*	<.001
Ansari et al.	Default	386	0.8506 *(0.8087–0.8896)*	0.5284 *(0.4553–0.5959)*	0.7435 *(0.7073–0.7798)*	0.4804 *(0.4078–0.5531)*	0.9710 *(0.9469–0.9903)*	0.9348 *(0.8824–0.9787)*	0.6347 *(0.5649–0.6975)*	<.001
HEPAD	Default	386	0.4918 *(0.4331–0.5510)*	0.0348 *(-0.0651–0.1365)*	0.5078 *(0.4585–0.5570)*	0.6425 *(0.5698–0.7095)*	0.3913 *(0.3237–0.4589)*	0.4772 *(0.4383–0.5168)*	0.5476 *(0.4988–0.5943)*	<.001
AMPDeep	Prot-BERT-BFD	386	0.6023 *(0.5456–0.6577)*	0.0398 *(-0.0618–0.1319)*	0.4767 *(0.4534–0.5000)*	0.9553 *(0.9218–0.9832)*	0.0628 *(0.0338–0.0966)*	0.4685 *(0.4566–0.4806)*	0.6287 *(0.6122–0.6435)*	<.001

TABLE S4

PERFORMANCE COMPARISON OF HLPPRED 2.0 WITH EXISTING PREDICTORS ON METHOD-SPECIFIC NON-OVERLAPPING INDEPENDENT TEST SUBSETS.

Method	Model	N	AUC	MCC	ACC	Sn	Sp	Precision	F1-score	P-value (MCC)
HemoPI2 (With MERCI)	ESM2-t6	386	0.9195 *(0.8918–0.9441)*	0.6515 *(0.5729–0.7289)*	0.8264 *(0.7876–0.8653)*	0.8212 *(0.7654–0.8771)*	0.8309 *(0.7778–0.8792)*	0.8077 *(0.7604–0.8571)*	0.8144 *(0.7718–0.8547)*	0.0040
	HLPpred 2.0 (same subset)	*386*	*0.9546 (0.9364–0.9723)*	*0.7654 (0.7028–0.8282)*	*0.8834 (0.8523–0.9145)*	*0.8659 (0.8156–0.9162)*	*0.8986 (0.8551–0.9372)*	*0.8807 (0.8387–0.9235)*	*0.8732 (0.8371–0.9071)*	—
HemoPI2 (With MERCI)	RF	386	0.9201 *(0.8934–0.9449)*	0.6732 *(0.5991–0.7459)*	0.8368 *(0.8005–0.8731)*	0.8436 *(0.7877–0.8939)*	0.8309 *(0.7778–0.8792)*	0.8118 *(0.7668–0.8611)*	0.8274 *(0.7875–0.8658)*	0.0142

Method	Model	N	AUC	MCC	ACC	Sn	Sp	Precision	F1-score	P-value (MCC)
	HLPpred 2.0 (same subset)	*386*	*0.9546 (0.9364–0.9723)*	*0.7654 (0.7028–0.8282)*	*0.8834 (0.8523–0.9145)*	*0.8659 (0.8156–0.9162)*	*0.8986 (0.8551–0.9372)*	*0.8807 (0.8387–0.9235)*	*0.8732 (0.8371–0.9071)*	—
HemoPI2 (Without MERCI)	ESM2-t6	386	0.8747 *(0.8377–0.9091)*	0.5895 *(0.5087–0.6702)*	0.7953 *(0.7539–0.8342)*	0.7933 *(0.7318–0.8492)*	0.7971 *(0.7391–0.8502)*	0.7717 *(0.7243–0.8232)*	0.7824 *(0.7380–0.8248)*	<.001
	HLPpred 2.0 (same subset)	*386*	*0.9546 (0.9364–0.9723)*	*0.7654 (0.7028–0.8282)*	*0.8834 (0.8523–0.9145)*	*0.8659 (0.8156–0.9162)*	*0.8986 (0.8551–0.9372)*	*0.8807 (0.8387–0.9235)*	*0.8732 (0.8371–0.9071)*	—
HemoPI2 (Without MERCI)	RF	386	0.8882 *(0.8546–0.9198)*	0.6168 *(0.5380–0.6945)*	0.8083 *(0.7694–0.8472)*	0.8212 *(0.7654–0.8771)*	0.7971 *(0.7440–0.8502)*	0.7778 *(0.7320–0.8280)*	0.7989 *(0.7569–0.8392)*	<.001
	HLPpred 2.0 (same subset)	*386*	*0.9546 (0.9364–0.9723)*	*0.7654 (0.7028–0.8282)*	*0.8834 (0.8523–0.9145)*	*0.8659 (0.8156–0.9162)*	*0.8986 (0.8551–0.9372)*	*0.8807 (0.8387–0.9235)*	*0.8732 (0.8371–0.9071)*	—
HLPpred-Fuse	ERT-based meta model	354	0.5543 *(0.4970–0.6105)*	-0.0088 *(-0.1127–0.0926)*	0.4379 *(0.4153–0.4633)*	0.9412 *(0.9020–0.9739)*	0.0547 *(0.0249–0.0896)*	0.4311 *(0.4182–0.4435)*	0.5914 *(0.5720–0.6085)*	<.001
	HLPpred 2.0 (same subset)	*354*	*0.9562 (0.9353–0.9740)*	*0.7642 (0.6943–0.8328)*	*0.8842 (0.8503–0.9181)*	*0.8693 (0.8170–0.9216)*	*0.8955 (0.8507–0.9353)*	*0.8636 (0.8153–0.9128)*	*0.8664 (0.8255–0.9043)*	—
HemoPI	SVM	334	0.6381 *(0.5759–0.6973)*	0.2334 *(0.1266–0.3378)*	0.6287 *(0.5838–0.6737)*	0.3581 *(0.2838–0.4324)*	0.8441 *(0.7903–0.8925)*	0.6463 *(0.5556–0.7375)*	0.4609 *(0.3779–0.5378)*	<.001
	HLPpred 2.0 (same subset)	*334*	*0.9584 (0.9365–0.9762)*	*0.7878 (0.7204–0.8543)*	*0.8952 (0.8623–0.9281)*	*0.8851 (0.8311–0.9324)*	*0.9032 (0.8602–0.9409)*	*0.8792 (0.8333–0.9252)*	*0.8822 (0.8439–0.9184)*	—
HemoPred	Default	302	0.6568 *(0.5926–0.7199)*	0.1353 *(0.0251–0.2440)*	0.5232 *(0.4735–0.5728)*	0.8047 *(0.7344–0.8672)*	0.3161 *(0.2471–0.3851)*	0.4640 *(0.4313–0.4977)*	0.5886 *(0.5465–0.6289)*	<.001
	HLPpred 2.0 (same subset)	*302*	*0.9581 (0.9347–0.9773)*	*0.7896 (0.7208–0.8575)*	*0.8974 (0.8642–0.9305)*	*0.8750 (0.8125–0.9297)*	*0.9138 (0.8678–0.9540)*	*0.8819 (0.8310–0.9333)*	*0.8784 (0.8367–0.9173)*	—
HemoNet	CNN-based	208	0.6581 *(0.5815–0.7306)*	0.2027 *(0.0661–0.3335)*	0.6058 *(0.5385–0.6731)*	0.5795 *(0.4773–0.6818)*	0.6250 *(0.5333–0.7083)*	0.5312 *(0.4574–0.6061)*	0.5543 *(0.4719–0.6292)*	<.001
	HLPpred 2.0 (same subset)	*208*	*0.9506 (0.9223–0.9748)*	*0.7535 (0.6633–0.8424)*	*0.8798 (0.8365–0.9231)*	*0.8523 (0.7727–0.9205)*	*0.9000 (0.8417–0.9500)*	*0.8621 (0.8000–0.9268)*	*0.8571 (0.8023–0.9080)*	—
Transformer-CNN	Single Architecture	275	0.5718 *(0.5030–0.6378)*	0.1250 *(0.0103–0.2367)*	0.5236 *(0.4727–0.5745)*	0.7881 *(0.7119–0.8559)*	0.3248 *(0.2548–0.4013)*	0.4673 *(0.4322–0.5027)*	0.5868 *(0.5415–0.6293)*	<.001
	HLPpred 2.0 (same subset)	*275*	*0.9612 (0.9404–0.9787)*	*0.7638 (0.6869–0.8378)*	*0.8836 (0.8436–0.9200)*	*0.8814 (0.8220–0.9322)*	*0.8854 (0.8344–0.9299)*	*0.8525 (0.7971–0.9076)*	*0.8667 (0.8226–0.9083)*	—
Transformer-CNN	Combined Architecture	275	0.5657 *(0.4966–0.6344)*	0.0365 *(-0.0799–0.1507)*	0.4618 *(0.4218–0.5018)*	0.8729 *(0.8136–0.9322)*	0.1529 *(0.0955–0.2102)*	0.4364 *(0.4132–0.4599)*	0.5819 *(0.5491–0.6127)*	<.001
	HLPpred 2.0 (same subset)	*275*	*0.9612 (0.9404–0.9787)*	*0.7638 (0.6869–0.8378)*	*0.8836 (0.8436–0.9200)*	*0.8814 (0.8220–0.9322)*	*0.8854 (0.8344–0.9299)*	*0.8525 (0.7971–0.9076)*	*0.8667 (0.8226–0.9083)*	—
PeptideBERT	Default	33	0.7610 *(0.5882–0.9154)*	0.3068 *(0.0000–0.4573)*	0.5758 *(0.4848–0.6667)*	0.1765 *(0.0000–0.3529)*	1.0000 *(1.0000–1.0000)*	1.0000 *(0.0000–1.0000)*	0.3000 *(0.0000–0.5217)*	0.0838
	HLPpred 2.0 (same subset)	*33*	*0.9228 (0.8199–0.9890)*	*0.5756 (0.2750–0.8310)*	*0.7879 (0.6364–0.9091)*	*0.8235 (0.6471–1.0000)*	*0.7500 (0.5000–0.9375)*	*0.7778 (0.6364–0.9375)*	*0.8000 (0.6471–0.9189)*	—
Ansari et al.	Default	33	0.8971 *(0.7721–0.9853)*	0.5034 *(0.3068–0.6860)*	0.6970 *(0.5758–0.8182)*	0.4118 *(0.1765–0.6471)*	1.0000 *(1.0000–1.0000)*	1.0000 *(1.0000–1.0000)*	0.5833 *(0.3000–0.7857)*	0.5998
	HLPpred 2.0 (same subset)	*33*	*0.9228 (0.8199–0.9890)*	*0.5756 (0.2750–0.8310)*	*0.7879 (0.6364–0.9091)*	*0.8235 (0.6471–1.0000)*	*0.7500 (0.5000–0.9375)*	*0.7778 (0.6364–0.9375)*	*0.8000 (0.6471–0.9189)*	—
HEPAD	Default	280	0.4977	0.0013	0.4714	0.6579	0.3434	0.4076	0.5034	<.001

Method	Model	N	AUC	MCC	ACC	Sn	Sp	Precision	F1-score	P-value (MCC)
			(0.4288–0.5661)	*(-0.1170–0.1170)*	*(0.4143–0.5286)*	*(0.5702–0.7456)*	*(0.2711–0.4157)*	*(0.3656–0.4492)*	*(0.4476–0.5548)*	
	HLPpred 2.0 (same subset)	*280*	*0.9579 (0.9338–0.9781)*	*0.7929 (0.7182–0.8610)*	*0.9000 (0.8643–0.9321)*	*0.8772 (0.8158–0.9386)*	*0.9157 (0.8735–0.9578)*	*0.8772 (0.8235–0.9304)*	*0.8772 (0.8319–0.9185)*	—
AMPDeep	Prot-BERT-BFD	316	0.5991 *(0.5346–0.6625)*	-0.0325 *(-0.1447–0.0757)*	0.4304 *(0.4051–0.4557)*	0.9412 *(0.8971–0.9779)*	0.0444 *(0.0167–0.0778)*	0.4267 *(0.4136–0.4389)*	0.5872 *(0.5674–0.6050)*	<.001
	HLPpred 2.0 (same subset)	*316*	*0.9548 (0.9326–0.9735)*	*0.7491 (0.6741–0.8202)*	*0.8766 (0.8386–0.9114)*	*0.8676 (0.8088–0.9265)*	*0.8833 (0.8333–0.9278)*	*0.8489 (0.7973–0.9023)*	*0.8582 (0.8148–0.8986)*	—

TABLE S5

INFERENCE TIME, THROUGHPUT, AND MEMORY USAGE OF HLPPRED 2.0 AT DIFFERENT BATCH SIZES

Task	No. of peptides	Total time (s)	Time/peptide (s)	Throughput (peptides/h)	Peak RAM (GB)	Peak GPU memory (GB)
Classification	1	335.29 ± 8.31	335.29 ± 8.31	11	21.00 ± 0.03	10.93 ± 0.88
	10	356.73 ± 3.05	35.67 ± 0.31	101	20.37 ± 0.14	11.72 ± 0.15
	50	357.00 ± 18.81	7.14 ± 0.38	504	21.17 ± 0.44	11.87 ± 0.00
	386	512.16 ± 13.20	**1.33 ± 0.03**	**2,713**	21.15 ± 0.53	11.87 ± 0.00
Regression	1	278.95 ± 10.44	278.95 ± 10.44	13	20.62 ± 0.30	10.88 ± 0.45
	10	300.46 ± 4.72	30.05 ± 0.47	120	20.60 ± 0.53	11.65 ± 0.37
	50	312.67 ± 12.43	6.25 ± 0.25	576	20.98 ± 0.36	11.87 ± 0.00
	386	440.81 ± 9.88	**1.14 ± 0.03**	**3,152**	21.61 ± 0.27	11.87 ± 0.00

TABLE S6

STAGE-WISE COMPUTATIONAL COST OF HLPPRED 2.0 INFERENCE OF A BATCH OF 386 PEPTIDES

Stage	Classification	Regression
MERCI motif scan	0.60 ± 0.02 s (0.1%)	N/A
Feature extraction	326.37 ± 7.58 s (63.7%)	331.39 ± 4.44 s (75.2%)
Meta-feature inference	153.62 ± 5.62 s (30.0%)	82.24 ± 4.45 s (18.7%)
SeqVec embedding	15.80 ± 0.31 s (3.1%)	15.93 ± 0.50 s (3.6%)
DL ensemble (5 folds)	14.07 ± 0.48 s (2.7%)	11.22 ± 1.17 s (2.5%)

TABLE S7

PERFORMANCE OF 360 BASELINE SINGLE FEATURE-MODELS ON THE TRAINING DATASET AND INDEPENDENT DATASET FOR THE CLASSIFICATION TASK

Classifier Models	Training							Independent						
	AUC	MCC	ACC	Sn	Sp	Pre	F1	AUC	MCC	ACC	Sn	Sp	Pre	F1
AB_Bepler	0.82	0.516	0.76	0.719	0.795	0.752	0.734	0.835	0.525	0.764	0.687	0.831	0.779	0.73
AB_CPCProt	0.811	0.502	0.75	0.697	0.801	0.752	0.722	0.854	0.563	0.782	0.698	0.855	0.807	0.749
AB_ESM	0.859	0.576	0.79	0.744	0.827	0.788	0.764	0.867	0.609	0.806	0.754	0.85	0.813	0.783
AB_ESM1b	0.858	0.569	0.79	0.733	0.83	0.789	0.758	0.874	0.583	0.793	0.721	0.855	0.811	0.763
AB_ESM1v	0.85	0.588	0.79	0.759	0.826	0.791	0.774	0.861	0.57	0.785	0.682	0.874	0.824	0.746
AB_ESM2-t6	0.865	0.598	0.8	0.76	0.833	0.799	0.777	0.864	0.562	0.782	0.726	0.831	0.788	0.756
AB_ESM2-t12	0.853	0.577	0.79	0.751	0.822	0.786	0.767	0.842	0.536	0.769	0.737	0.797	0.759	0.748
AB_ESM2-t30	0.851	0.567	0.78	0.737	0.824	0.784	0.758	0.859	0.551	0.777	0.715	0.831	0.785	0.749
AB_ESM2-t33	0.868	0.586	0.79	0.751	0.831	0.794	0.771	0.886	0.656	0.829	0.777	0.874	0.842	0.808
AB_ESM2-t36	0.854	0.566	0.78	0.718	0.841	0.797	0.754	0.884	0.604	0.803	0.737	0.86	0.82	0.777
AB_FastText	0.838	0.563	0.78	0.732	0.826	0.786	0.757	0.846	0.552	0.777	0.698	0.845	0.796	0.744
AB_GloVe	0.819	0.52	0.76	0.725	0.792	0.751	0.737	0.823	0.504	0.754	0.71	0.792	0.747	0.728
AB_PRNN	0.822	0.535	0.77	0.718	0.813	0.769	0.742	0.887	0.647	0.824	0.754	0.884	0.849	0.799

AB_PTAB	0.843	0.569	0.79	0.73	0.832	0.791	0.758	0.834	0.541	0.772	0.698	0.836	0.786	0.74
AB_PTBB	0.829	0.515	0.76	0.722	0.79	0.749	0.734	0.808	0.473	0.738	0.704	0.768	0.724	0.714
AB_PTB	0.852	0.592	0.8	0.754	0.833	0.796	0.774	0.863	0.6	0.801	0.721	0.87	0.827	0.77
AB_PTU	0.864	0.603	0.8	0.765	0.835	0.8	0.781	0.849	0.572	0.788	0.737	0.831	0.79	0.763
AB_PTXU	0.865	0.596	0.8	0.764	0.828	0.794	0.778	0.868	0.609	0.806	0.743	0.86	0.821	0.78
AB_PTXLU	0.799	0.463	0.73	0.684	0.775	0.725	0.703	0.82	0.483	0.744	0.693	0.787	0.738	0.715
AB_Seq2Vec	0.871	0.607	0.8	0.766	0.838	0.804	0.783	0.888	0.62	0.811	0.749	0.865	0.827	0.786
AB_Word2Vec	0.834	0.557	0.78	0.737	0.816	0.777	0.756	0.851	0.562	0.782	0.726	0.831	0.788	0.756
AB_AAC	0.801	0.484	0.74	0.691	0.787	0.739	0.712	0.815	0.504	0.754	0.715	0.787	0.744	0.729
AB_APAAC	0.824	0.532	0.77	0.708	0.819	0.772	0.737	0.853	0.588	0.795	0.76	0.826	0.791	0.775
AB_ASDC	0.812	0.488	0.74	0.711	0.774	0.733	0.72	0.847	0.562	0.782	0.754	0.807	0.771	0.763
AB_CKSAAGP-type-1	0.785	0.453	0.73	0.688	0.762	0.715	0.7	0.824	0.475	0.738	0.626	0.836	0.767	0.689
AB_CKSAAGP-type-2	0.797	0.477	0.74	0.695	0.778	0.731	0.711	0.825	0.501	0.751	0.648	0.841	0.779	0.707
AB_CKSAAP-type-1	0.817	0.504	0.75	0.709	0.791	0.747	0.726	0.842	0.541	0.772	0.704	0.831	0.783	0.741
AB_CKSAAP-type-2	0.828	0.512	0.76	0.709	0.8	0.753	0.729	0.827	0.557	0.78	0.721	0.831	0.787	0.752
AB_CTDC	0.803	0.48	0.74	0.697	0.779	0.733	0.713	0.817	0.472	0.738	0.676	0.792	0.738	0.706
AB_CTDD	0.824	0.521	0.76	0.706	0.809	0.763	0.732	0.846	0.525	0.764	0.687	0.831	0.779	0.73
AB_CTDT	0.787	0.44	0.72	0.677	0.761	0.71	0.692	0.793	0.436	0.72	0.654	0.778	0.718	0.684
AB_CTriad	0.766	0.417	0.71	0.655	0.758	0.7	0.675	0.786	0.462	0.733	0.665	0.792	0.735	0.698
AB_DDE	0.821	0.516	0.76	0.729	0.785	0.747	0.736	0.847	0.583	0.793	0.76	0.821	0.786	0.773
AB_DistancePair	0.801	0.484	0.74	0.691	0.787	0.739	0.712	0.815	0.504	0.754	0.715	0.787	0.744	0.729
AB_DPC-type-1	0.808	0.475	0.74	0.691	0.78	0.731	0.709	0.809	0.514	0.759	0.71	0.802	0.756	0.732
AB_DPC-type-2	0.807	0.504	0.75	0.692	0.807	0.754	0.721	0.802	0.484	0.744	0.648	0.826	0.763	0.701
AB_GDPC-type-2	0.76	0.419	0.71	0.658	0.757	0.701	0.677	0.778	0.394	0.7	0.603	0.783	0.706	0.651
AB_PAAC	0.826	0.536	0.77	0.716	0.815	0.77	0.741	0.854	0.557	0.78	0.71	0.841	0.794	0.749
AB_PDE	0.845	0.569	0.79	0.744	0.821	0.783	0.762	0.865	0.614	0.808	0.782	0.831	0.8	0.791
AB_QSOrder	0.827	0.525	0.76	0.716	0.804	0.761	0.737	0.843	0.577	0.79	0.743	0.831	0.792	0.767
ANN_Bepler	0.805	0.495	0.75	0.75	0.744	0.718	0.732	0.836	0.528	0.759	0.587	0.908	0.847	0.693
ANN_CPCProt	0.797	0.464	0.73	0.706	0.756	0.715	0.71	0.806	0.492	0.746	0.749	0.744	0.717	0.732
ANN_ESM	0.849	0.588	0.79	0.764	0.82	0.788	0.774	0.871	0.572	0.788	0.754	0.816	0.78	0.767
ANN_ESM1b	0.841	0.55	0.77	0.751	0.795	0.761	0.754	0.86	0.557	0.78	0.765	0.792	0.761	0.763
ANN_ESM1v	0.844	0.549	0.78	0.751	0.796	0.762	0.756	0.84	0.504	0.754	0.676	0.821	0.766	0.718
ANN_ESM2-t6	0.853	0.587	0.79	0.782	0.803	0.775	0.777	0.825	0.494	0.746	0.771	0.725	0.708	0.738
ANN_ESM2-t12	0.842	0.559	0.78	0.759	0.798	0.765	0.76	0.821	0.489	0.746	0.721	0.768	0.729	0.725
ANN_ESM2-t30	0.841	0.559	0.78	0.74	0.814	0.776	0.756	0.833	0.559	0.78	0.682	0.865	0.813	0.742
ANN_ESM2-t33	0.847	0.572	0.79	0.77	0.8	0.77	0.768	0.87	0.578	0.79	0.726	0.845	0.803	0.763
ANN_ESM2-t36	0.836	0.552	0.78	0.749	0.802	0.765	0.756	0.865	0.552	0.777	0.754	0.797	0.763	0.758
ANN_FastText	0.84	0.55	0.78	0.742	0.806	0.767	0.753	0.851	0.557	0.78	0.743	0.812	0.773	0.758
ANN_GloVe	0.817	0.496	0.75	0.727	0.767	0.732	0.729	0.815	0.483	0.744	0.676	0.802	0.747	0.71
ANN_PRNN	0.813	0.501	0.75	0.743	0.755	0.727	0.732	0.835	0.483	0.744	0.687	0.792	0.741	0.713
ANN_PTAB	0.83	0.54	0.77	0.754	0.783	0.751	0.75	0.804	0.436	0.72	0.643	0.787	0.723	0.681
ANN_PTBB	0.804	0.491	0.75	0.739	0.752	0.719	0.728	0.773	0.374	0.687	0.704	0.672	0.65	0.676
ANN_PTB	0.849	0.563	0.78	0.764	0.797	0.766	0.764	0.854	0.551	0.777	0.726	0.821	0.778	0.751
ANN_PTU	0.855	0.581	0.79	0.76	0.818	0.784	0.77	0.8	0.457	0.731	0.648	0.802	0.739	0.691
ANN_PTXU	0.854	0.581	0.79	0.765	0.814	0.781	0.773	0.838	0.509	0.757	0.71	0.797	0.752	0.73
ANN_PTXLU	0.782	0.449	0.72	0.711	0.736	0.7	0.703	0.757	0.389	0.694	0.704	0.686	0.66	0.681
ANN_Seq2Vec	0.866	0.607	0.8	0.802	0.803	0.779	0.789	0.862	0.593	0.798	0.754	0.836	0.799	0.776
ANN_Word2Vec	0.822	0.527	0.76	0.74	0.783	0.749	0.742	0.838	0.531	0.767	0.743	0.787	0.751	0.747
ANN_AAC	0.781	0.43	0.72	0.685	0.742	0.699	0.69	0.788	0.467	0.733	0.749	0.72	0.698	0.722
ANN_APAAC	0.826	0.538	0.77	0.756	0.78	0.75	0.751	0.815	0.489	0.746	0.654	0.826	0.765	0.705
ANN_ASDC	0.776	0.436	0.72	0.676	0.757	0.707	0.69	0.787	0.473	0.738	0.693	0.778	0.729	0.711
ANN_CKSAAGP-type-1	0.767	0.415	0.71	0.669	0.742	0.692	0.675	0.781	0.457	0.731	0.648	0.802	0.739	0.691
ANN_CKSAAGP-type-2	0.771	0.444	0.72	0.698	0.743	0.703	0.697	0.812	0.476	0.738	0.743	0.734	0.707	0.725
ANN_CKSAAP-type-1	0.78	0.426	0.71	0.684	0.739	0.695	0.688	0.782	0.411	0.707	0.682	0.73	0.685	0.684
ANN_CKSAAP-type-2	0.811	0.513	0.76	0.72	0.79	0.749	0.733	0.806	0.478	0.741	0.67	0.802	0.745	0.706
ANN_CTDC	0.777	0.438	0.72	0.691	0.744	0.701	0.694	0.779	0.42	0.71	0.721	0.701	0.675	0.697
ANN_CTDD	0.79	0.48	0.74	0.688	0.786	0.737	0.71	0.781	0.441	0.723	0.665	0.773	0.717	0.69

ANN_CTDT	0.755	0.392	0.7	0.691	0.7	0.667	0.677	0.739	0.399	0.702	0.609	0.783	0.708	0.655
ANN_CTriad	0.726	0.376	0.69	0.632	0.74	0.677	0.651	0.732	0.321	0.658	0.698	0.623	0.616	0.655
ANN_DDE	0.802	0.487	0.74	0.719	0.766	0.726	0.721	0.777	0.481	0.741	0.62	0.845	0.776	0.689
ANN_DistancePair	0.781	0.43	0.72	0.685	0.742	0.699	0.69	0.788	0.467	0.733	0.749	0.72	0.698	0.722
ANN_DPC-type-1	0.767	0.417	0.71	0.66	0.752	0.699	0.677	0.736	0.362	0.684	0.603	0.754	0.679	0.639
ANN_DPC-type-2	0.805	0.496	0.75	0.727	0.766	0.729	0.726	0.81	0.517	0.759	0.76	0.759	0.731	0.745
ANN_GDPC-type-2	0.763	0.41	0.71	0.677	0.732	0.685	0.68	0.762	0.416	0.71	0.676	0.739	0.691	0.684
ANN_PAAC	0.835	0.52	0.76	0.732	0.784	0.749	0.737	0.835	0.554	0.777	0.782	0.773	0.749	0.765
ANN_PDE	0.824	0.511	0.76	0.73	0.779	0.74	0.734	0.845	0.521	0.762	0.737	0.783	0.746	0.742
ANN_QSOrder	0.823	0.533	0.77	0.758	0.774	0.744	0.75	0.824	0.511	0.757	0.743	0.768	0.735	0.739
CB_Bepler	0.791	0.468	0.74	0.681	0.783	0.731	0.704	0.809	0.457	0.731	0.665	0.787	0.73	0.696
CB_CPCProt	0.784	0.459	0.73	0.691	0.766	0.718	0.704	0.831	0.535	0.769	0.715	0.816	0.771	0.742
CB_ESM	0.837	0.532	0.77	0.723	0.806	0.762	0.741	0.831	0.499	0.751	0.67	0.821	0.764	0.714
CB_ESM1b	0.829	0.53	0.77	0.702	0.821	0.773	0.734	0.83	0.525	0.764	0.721	0.802	0.759	0.739
CB_ESM1v	0.828	0.531	0.77	0.732	0.796	0.758	0.743	0.816	0.515	0.759	0.67	0.836	0.779	0.721
CB_ESM2-t6	0.835	0.556	0.78	0.753	0.802	0.766	0.759	0.807	0.452	0.728	0.693	0.759	0.713	0.703
CB_ESM2-t12	0.817	0.511	0.76	0.72	0.787	0.746	0.732	0.82	0.5	0.751	0.726	0.773	0.735	0.73
CB_ESM2-t30	0.826	0.519	0.76	0.725	0.792	0.751	0.737	0.823	0.499	0.751	0.698	0.797	0.749	0.723
CB_ESM2-t33	0.84	0.55	0.78	0.744	0.803	0.766	0.754	0.859	0.567	0.785	0.743	0.821	0.782	0.762
CB_ESM2-t36	0.819	0.506	0.75	0.698	0.803	0.756	0.724	0.853	0.546	0.775	0.721	0.821	0.777	0.748
CB_FastText	0.808	0.496	0.75	0.704	0.789	0.743	0.722	0.802	0.436	0.72	0.654	0.778	0.718	0.684
CB_GloVe	0.78	0.456	0.73	0.68	0.772	0.721	0.698	0.781	0.39	0.697	0.659	0.73	0.678	0.669
CB_PRNN	0.805	0.489	0.75	0.711	0.775	0.732	0.72	0.865	0.572	0.788	0.754	0.816	0.78	0.767
CB_PTAB	0.827	0.517	0.76	0.721	0.793	0.752	0.735	0.795	0.478	0.741	0.682	0.792	0.739	0.709
CB_PTBB	0.788	0.466	0.73	0.708	0.757	0.714	0.71	0.766	0.468	0.736	0.704	0.763	0.72	0.712
CB_PTB	0.831	0.518	0.76	0.735	0.781	0.743	0.738	0.846	0.557	0.78	0.749	0.807	0.77	0.759
CB_PTU	0.84	0.539	0.77	0.733	0.802	0.763	0.746	0.803	0.468	0.736	0.698	0.768	0.723	0.71
CB_PTXU	0.839	0.53	0.77	0.747	0.781	0.747	0.746	0.835	0.515	0.759	0.715	0.797	0.753	0.734
CB_PTXLU	0.769	0.415	0.71	0.676	0.738	0.69	0.682	0.772	0.463	0.733	0.704	0.759	0.716	0.71
CB_Seq2Vec	0.841	0.558	0.78	0.751	0.804	0.769	0.759	0.833	0.572	0.788	0.737	0.831	0.79	0.763
CB_Word2Vec	0.798	0.472	0.74	0.698	0.772	0.725	0.71	0.827	0.515	0.759	0.732	0.783	0.744	0.738
CB_AAC	0.804	0.481	0.74	0.714	0.766	0.725	0.718	0.825	0.504	0.754	0.721	0.783	0.741	0.731
CB_APAAC	0.822	0.516	0.76	0.711	0.801	0.756	0.731	0.823	0.473	0.738	0.665	0.802	0.744	0.702
CB_ASDC	0.8	0.484	0.74	0.715	0.767	0.726	0.719	0.848	0.556	0.78	0.737	0.816	0.777	0.756
CB_CKSAAGP-type-1	0.789	0.473	0.74	0.698	0.772	0.725	0.71	0.825	0.486	0.744	0.626	0.845	0.778	0.694
CB_CKSAAGP-type-2	0.8	0.472	0.74	0.691	0.777	0.729	0.708	0.838	0.552	0.777	0.698	0.845	0.796	0.744
CB_CKSAAP-type-1	0.816	0.515	0.76	0.695	0.813	0.765	0.726	0.816	0.478	0.741	0.659	0.812	0.752	0.702
CB_CKSAAP-type-2	0.822	0.513	0.76	0.695	0.812	0.763	0.726	0.836	0.473	0.738	0.665	0.802	0.744	0.702
CB_CTDC	0.796	0.47	0.74	0.701	0.766	0.724	0.71	0.812	0.485	0.744	0.637	0.836	0.77	0.697
CB_CTDD	0.815	0.498	0.75	0.706	0.787	0.744	0.723	0.845	0.499	0.751	0.665	0.826	0.768	0.713
CB_CTDT	0.784	0.44	0.72	0.673	0.765	0.712	0.691	0.798	0.441	0.723	0.654	0.783	0.722	0.686
CB_CTriad	0.76	0.435	0.72	0.671	0.761	0.709	0.689	0.807	0.484	0.744	0.654	0.821	0.76	0.703
CB_DDE	0.822	0.528	0.77	0.719	0.806	0.761	0.739	0.848	0.532	0.767	0.754	0.778	0.746	0.75
CB_DistancePair	0.804	0.481	0.74	0.714	0.766	0.725	0.718	0.825	0.504	0.754	0.721	0.783	0.741	0.731
CB_DPC-type-1	0.809	0.483	0.74	0.711	0.771	0.728	0.718	0.82	0.514	0.759	0.704	0.807	0.759	0.73
CB_DPC-type-2	0.801	0.492	0.75	0.701	0.787	0.74	0.719	0.826	0.499	0.751	0.682	0.812	0.758	0.718
CB_GDPC-type-2	0.778	0.431	0.72	0.665	0.763	0.707	0.684	0.848	0.522	0.762	0.659	0.85	0.792	0.72
CB_PAAC	0.819	0.521	0.76	0.707	0.809	0.763	0.733	0.843	0.505	0.754	0.67	0.826	0.769	0.716
CB_PDE	0.841	0.558	0.78	0.739	0.816	0.776	0.756	0.875	0.593	0.798	0.76	0.831	0.795	0.777
CB_QSOrder	0.825	0.511	0.76	0.715	0.792	0.75	0.73	0.853	0.572	0.788	0.76	0.812	0.777	0.768
ERT_Bepler	0.801	0.475	0.74	0.66	0.807	0.748	0.7	0.824	0.48	0.741	0.626	0.841	0.772	0.691
ERT_CPCProt	0.797	0.502	0.75	0.673	0.82	0.765	0.714	0.854	0.517	0.759	0.648	0.855	0.795	0.714
ERT_ESM	0.84	0.551	0.78	0.68	0.857	0.806	0.735	0.835	0.55	0.775	0.659	0.874	0.819	0.731
ERT_ESM1b	0.831	0.538	0.77	0.667	0.858	0.802	0.727	0.846	0.5	0.751	0.659	0.831	0.771	0.711
ERT_ESM1v	0.842	0.541	0.77	0.687	0.844	0.792	0.735	0.848	0.515	0.757	0.626	0.87	0.806	0.704
ERT_ESM2-t6	0.845	0.565	0.78	0.706	0.849	0.803	0.75	0.844	0.531	0.767	0.687	0.836	0.783	0.732
ERT_ESM2-t12	0.832	0.537	0.77	0.677	0.849	0.795	0.73	0.842	0.548	0.775	0.676	0.86	0.807	0.736

ERT_ESM2-t30	0.83	0.531	0.77	0.681	0.839	0.786	0.728	0.83	0.519	0.759	0.637	0.865	0.803	0.71
ERT_ESM2-t33	0.848	0.554	0.78	0.69	0.853	0.802	0.74	0.86	0.603	0.801	0.693	0.894	0.849	0.763
ERT_ESM2-t36	0.827	0.532	0.77	0.666	0.851	0.797	0.723	0.847	0.541	0.769	0.643	0.879	0.821	0.721
ERT_FastText	0.814	0.522	0.76	0.664	0.845	0.789	0.72	0.825	0.508	0.754	0.631	0.86	0.796	0.704
ERT_GloVe	0.79	0.469	0.74	0.631	0.826	0.758	0.688	0.801	0.463	0.733	0.626	0.826	0.757	0.685
ERT_PRNN	0.818	0.51	0.76	0.65	0.845	0.786	0.71	0.866	0.55	0.775	0.659	0.874	0.819	0.731
ERT_PTAB	0.831	0.547	0.77	0.68	0.854	0.803	0.735	0.839	0.539	0.769	0.659	0.865	0.808	0.726
ERT_PTBB	0.799	0.492	0.75	0.642	0.837	0.773	0.7	0.794	0.448	0.725	0.609	0.826	0.752	0.673
ERT_PTB	0.84	0.547	0.77	0.705	0.835	0.787	0.742	0.838	0.552	0.777	0.698	0.845	0.796	0.744
ERT_PTU	0.836	0.552	0.78	0.699	0.843	0.796	0.743	0.82	0.505	0.754	0.665	0.831	0.773	0.715
ERT_PTXU	0.842	0.554	0.78	0.712	0.835	0.789	0.747	0.843	0.547	0.775	0.687	0.85	0.799	0.739
ERT_PTXLU	0.785	0.44	0.72	0.635	0.797	0.731	0.678	0.798	0.473	0.738	0.648	0.816	0.753	0.697
ERT_Seq2Vec	0.85	0.567	0.78	0.695	0.86	0.812	0.747	0.871	0.563	0.78	0.654	0.889	0.836	0.734
ERT_Word2Vec	0.803	0.491	0.75	0.642	0.837	0.773	0.701	0.811	0.442	0.723	0.609	0.821	0.747	0.671
ERT_AAC	0.814	0.489	0.75	0.678	0.804	0.751	0.711	0.838	0.504	0.754	0.71	0.792	0.747	0.728
ERT_APAAC	0.831	0.537	0.77	0.695	0.833	0.784	0.736	0.851	0.568	0.785	0.71	0.85	0.804	0.754
ERT_ASDC	0.813	0.508	0.76	0.698	0.804	0.758	0.725	0.849	0.499	0.751	0.693	0.802	0.752	0.721
ERT_CKSAAGP-type-1	0.795	0.491	0.75	0.708	0.781	0.736	0.721	0.841	0.51	0.757	0.682	0.821	0.767	0.722
ERT_CKSAAGP-type-2	0.805	0.487	0.74	0.708	0.777	0.732	0.719	0.849	0.52	0.762	0.704	0.812	0.764	0.733
ERT_CKSAAP-type-1	0.817	0.509	0.76	0.684	0.818	0.764	0.721	0.839	0.483	0.744	0.67	0.807	0.75	0.708
ERT_CKSAAP-type-2	0.828	0.523	0.76	0.674	0.838	0.783	0.723	0.844	0.489	0.746	0.659	0.821	0.761	0.707
ERT_CTDC	0.804	0.482	0.74	0.685	0.792	0.74	0.711	0.818	0.489	0.746	0.654	0.826	0.765	0.705
ERT_CTDD	0.818	0.512	0.76	0.688	0.818	0.765	0.724	0.85	0.506	0.754	0.654	0.841	0.78	0.711
ERT_CTDT	0.787	0.444	0.72	0.66	0.779	0.721	0.689	0.793	0.43	0.718	0.637	0.787	0.722	0.677
ERT_CTriad	0.782	0.435	0.72	0.635	0.792	0.726	0.675	0.816	0.498	0.749	0.62	0.86	0.793	0.696
ERT_DDE	0.817	0.504	0.75	0.706	0.795	0.748	0.726	0.855	0.52	0.762	0.715	0.802	0.757	0.736
ERT_DistancePair	0.814	0.489	0.75	0.678	0.804	0.751	0.711	0.838	0.504	0.754	0.71	0.792	0.747	0.728
ERT_DPC-type-1	0.811	0.5	0.75	0.673	0.819	0.763	0.713	0.837	0.489	0.746	0.665	0.816	0.758	0.708
ERT_DPC-type-2	0.819	0.504	0.75	0.664	0.83	0.77	0.712	0.834	0.51	0.757	0.665	0.836	0.778	0.717
ERT_GDPC-type-2	0.778	0.423	0.71	0.659	0.761	0.704	0.68	0.823	0.473	0.738	0.654	0.812	0.75	0.699
ERT_PAAC	0.832	0.533	0.77	0.697	0.829	0.779	0.734	0.842	0.52	0.762	0.687	0.826	0.774	0.728
ERT_PDE	0.842	0.56	0.78	0.742	0.815	0.776	0.758	0.878	0.593	0.798	0.749	0.841	0.802	0.775
ERT_QSOrder	0.831	0.518	0.76	0.688	0.822	0.771	0.726	0.857	0.546	0.775	0.726	0.816	0.774	0.749
GB_Bepler	0.802	0.496	0.75	0.683	0.807	0.754	0.715	0.819	0.484	0.744	0.643	0.831	0.767	0.699
GB_CPCProt	0.8	0.501	0.75	0.685	0.809	0.756	0.718	0.837	0.522	0.762	0.659	0.85	0.792	0.72
GB_ESM	0.852	0.569	0.79	0.723	0.838	0.795	0.756	0.854	0.574	0.788	0.704	0.86	0.813	0.755
GB_ESM1b	0.844	0.561	0.78	0.712	0.841	0.794	0.749	0.851	0.541	0.772	0.704	0.831	0.783	0.741
GB_ESM1v	0.856	0.58	0.79	0.736	0.838	0.798	0.765	0.85	0.474	0.738	0.631	0.831	0.764	0.691
GB_ESM2-t6	0.857	0.589	0.79	0.73	0.85	0.811	0.766	0.85	0.526	0.764	0.676	0.841	0.786	0.727
GB_ESM2-t12	0.838	0.562	0.78	0.729	0.829	0.786	0.755	0.841	0.525	0.764	0.698	0.821	0.772	0.733
GB_ESM2-t30	0.844	0.55	0.78	0.711	0.832	0.786	0.745	0.84	0.489	0.746	0.654	0.826	0.765	0.705
GB_ESM2-t33	0.863	0.587	0.79	0.743	0.838	0.8	0.769	0.876	0.61	0.806	0.732	0.87	0.829	0.777
GB_ESM2-t36	0.844	0.554	0.78	0.699	0.844	0.797	0.743	0.863	0.563	0.782	0.693	0.86	0.811	0.747
GB_FastText	0.831	0.536	0.77	0.69	0.837	0.787	0.734	0.836	0.531	0.767	0.687	0.836	0.783	0.732
GB_GloVe	0.802	0.499	0.75	0.662	0.827	0.769	0.71	0.81	0.485	0.744	0.637	0.836	0.77	0.697
GB_PRNN	0.828	0.531	0.77	0.69	0.833	0.781	0.732	0.88	0.592	0.795	0.693	0.884	0.838	0.758
GB_PTAB	0.846	0.564	0.78	0.716	0.839	0.795	0.752	0.847	0.56	0.78	0.676	0.87	0.818	0.74
GB_PTBB	0.815	0.514	0.76	0.691	0.816	0.766	0.725	0.808	0.472	0.738	0.682	0.787	0.735	0.707
GB_PTB	0.853	0.575	0.79	0.733	0.837	0.795	0.762	0.842	0.557	0.78	0.71	0.841	0.794	0.749
GB_PTU	0.859	0.592	0.8	0.736	0.849	0.809	0.769	0.844	0.53	0.767	0.698	0.826	0.776	0.735
GB_PTXU	0.858	0.568	0.79	0.725	0.837	0.794	0.756	0.855	0.526	0.764	0.676	0.841	0.786	0.727
GB_PTXLU	0.792	0.452	0.73	0.646	0.797	0.735	0.685	0.818	0.468	0.736	0.643	0.816	0.752	0.693
GB_Seq2Vec	0.864	0.589	0.8	0.736	0.847	0.807	0.768	0.875	0.586	0.793	0.693	0.879	0.832	0.756
GB_Word2Vec	0.813	0.511	0.76	0.671	0.83	0.775	0.718	0.821	0.515	0.759	0.682	0.826	0.772	0.724
GB_AAC	0.805	0.477	0.74	0.677	0.795	0.741	0.706	0.838	0.514	0.759	0.71	0.802	0.756	0.732
GB_APAAC	0.816	0.506	0.75	0.698	0.803	0.755	0.724	0.849	0.546	0.775	0.71	0.831	0.784	0.745
GB_ASDC	0.822	0.516	0.76	0.716	0.796	0.753	0.733	0.858	0.546	0.775	0.732	0.812	0.771	0.751

GB_CKSAAGP-type-1	0.798	0.491	0.75	0.705	0.783	0.738	0.72	0.843	0.499	0.751	0.676	0.816	0.761	0.716
GB_CKSAAGP-type-2	0.804	0.492	0.75	0.699	0.788	0.741	0.718	0.836	0.531	0.767	0.687	0.836	0.783	0.732
GB_CKSAAP-type-1	0.82	0.526	0.76	0.715	0.807	0.763	0.737	0.854	0.515	0.759	0.687	0.821	0.769	0.726
GB_CKSAAP-type-2	0.833	0.543	0.77	0.695	0.839	0.789	0.738	0.854	0.531	0.767	0.693	0.831	0.78	0.734
GB_CTDC	0.799	0.475	0.74	0.688	0.783	0.734	0.708	0.814	0.483	0.744	0.659	0.816	0.756	0.705
GB_CTDD	0.821	0.518	0.76	0.699	0.813	0.763	0.729	0.858	0.549	0.775	0.67	0.865	0.811	0.734
GB_CTDT	0.788	0.457	0.73	0.68	0.773	0.724	0.7	0.809	0.483	0.744	0.676	0.802	0.747	0.71
GB_CTriad	0.785	0.447	0.73	0.625	0.812	0.741	0.677	0.814	0.499	0.749	0.615	0.865	0.797	0.694
GB_DDE	0.817	0.509	0.76	0.678	0.823	0.768	0.719	0.852	0.521	0.762	0.67	0.841	0.784	0.723
GB_DistancePair	0.805	0.477	0.74	0.677	0.795	0.741	0.706	0.838	0.514	0.759	0.71	0.802	0.756	0.732
GB_DPC-type-1	0.817	0.509	0.76	0.694	0.809	0.759	0.724	0.856	0.541	0.772	0.715	0.821	0.776	0.744
GB_DPC-type-2	0.823	0.513	0.76	0.699	0.808	0.759	0.726	0.844	0.572	0.788	0.737	0.831	0.79	0.763
GB_GDPC-type-2	0.77	0.413	0.71	0.669	0.743	0.691	0.679	0.838	0.515	0.759	0.693	0.816	0.765	0.727
GB_PAAC	0.821	0.507	0.75	0.669	0.828	0.771	0.715	0.82	0.463	0.733	0.631	0.821	0.753	0.687
GB_PDE	0.833	0.553	0.78	0.747	0.803	0.766	0.756	0.863	0.614	0.808	0.771	0.841	0.807	0.789
GB_QSOrder	0.829	0.515	0.76	0.68	0.825	0.774	0.72	0.847	0.504	0.754	0.676	0.821	0.766	0.718
LGB_Bepler	0.795	0.495	0.75	0.694	0.797	0.747	0.719	0.823	0.494	0.749	0.665	0.821	0.763	0.71
LGB_CPCProt	0.804	0.49	0.75	0.682	0.802	0.748	0.713	0.859	0.562	0.782	0.715	0.841	0.795	0.753
LGB_ESM	0.833	0.537	0.77	0.713	0.818	0.772	0.74	0.851	0.604	0.803	0.743	0.855	0.816	0.778
LGB_ESM1b	0.832	0.517	0.76	0.701	0.809	0.764	0.729	0.85	0.547	0.775	0.687	0.85	0.799	0.739
LGB_ESM1v	0.827	0.533	0.77	0.714	0.814	0.769	0.739	0.822	0.478	0.741	0.654	0.816	0.755	0.701
LGB_ESM2-t6	0.848	0.562	0.78	0.729	0.827	0.786	0.755	0.845	0.563	0.782	0.698	0.855	0.807	0.749
LGB_ESM2-t12	0.826	0.529	0.77	0.702	0.82	0.774	0.735	0.827	0.525	0.764	0.698	0.821	0.772	0.733
LGB_ESM2-t30	0.821	0.507	0.75	0.709	0.794	0.747	0.726	0.833	0.551	0.777	0.715	0.831	0.785	0.749
LGB_ESM2-t33	0.836	0.533	0.77	0.698	0.828	0.778	0.736	0.857	0.565	0.782	0.682	0.87	0.819	0.744
LGB_ESM2-t36	0.817	0.5	0.75	0.645	0.842	0.78	0.704	0.845	0.575	0.788	0.687	0.874	0.826	0.75
LGB_FastText	0.822	0.514	0.76	0.698	0.81	0.762	0.727	0.838	0.536	0.769	0.693	0.836	0.785	0.736
LGB_GloVe	0.804	0.49	0.75	0.681	0.803	0.749	0.712	0.807	0.457	0.731	0.676	0.778	0.725	0.699
LGB_PRNN	0.824	0.546	0.77	0.721	0.82	0.777	0.747	0.863	0.605	0.803	0.732	0.865	0.824	0.775
LGB_PTAB	0.797	0.486	0.74	0.64	0.832	0.769	0.697	0.803	0.426	0.715	0.609	0.807	0.732	0.665
LGB_PTBB	0.792	0.485	0.74	0.678	0.801	0.748	0.711	0.774	0.441	0.723	0.67	0.768	0.714	0.692
LGB_PTB	0.81	0.498	0.75	0.692	0.801	0.751	0.719	0.823	0.512	0.757	0.648	0.85	0.789	0.712
LGB_PTU	0.828	0.517	0.76	0.692	0.818	0.768	0.726	0.836	0.525	0.764	0.687	0.831	0.779	0.73
LGB_PTXU	0.834	0.538	0.77	0.719	0.813	0.771	0.741	0.831	0.521	0.762	0.67	0.841	0.784	0.723
LGB_PTXLU	0.766	0.408	0.71	0.641	0.763	0.7	0.668	0.778	0.399	0.702	0.631	0.763	0.698	0.663
LGB_Seq2Vec	0.838	0.551	0.78	0.719	0.826	0.783	0.749	0.869	0.6	0.801	0.715	0.874	0.831	0.769
LGB_Word2Vec	0.807	0.48	0.74	0.674	0.799	0.745	0.707	0.806	0.452	0.728	0.648	0.797	0.734	0.688
LGB_AAC	0.802	0.463	0.73	0.67	0.787	0.733	0.699	0.827	0.483	0.744	0.704	0.778	0.733	0.718
LGB_APAAC	0.804	0.49	0.75	0.679	0.804	0.752	0.712	0.836	0.536	0.769	0.698	0.831	0.781	0.738
LGB_ASDC	0.808	0.499	0.75	0.686	0.808	0.755	0.718	0.847	0.514	0.759	0.698	0.812	0.762	0.729
LGB_CKSAAGP-type-1	0.776	0.426	0.71	0.649	0.772	0.711	0.677	0.835	0.468	0.736	0.637	0.821	0.755	0.691
LGB_CKSAAGP-type-2	0.798	0.46	0.73	0.678	0.778	0.725	0.699	0.837	0.485	0.744	0.637	0.836	0.77	0.697
LGB_CKSAAP-type-1	0.803	0.464	0.73	0.649	0.806	0.747	0.692	0.81	0.432	0.718	0.603	0.816	0.74	0.665
LGB_CKSAAP-type-2	0.816	0.493	0.75	0.66	0.824	0.763	0.707	0.822	0.481	0.741	0.62	0.845	0.776	0.689
LGB_CTDC	0.792	0.432	0.72	0.656	0.771	0.714	0.682	0.813	0.467	0.736	0.659	0.802	0.742	0.698
LGB_CTDD	0.809	0.491	0.75	0.698	0.79	0.741	0.718	0.837	0.499	0.751	0.682	0.812	0.758	0.718
LGB_CTDT	0.778	0.41	0.71	0.654	0.752	0.695	0.673	0.785	0.462	0.733	0.67	0.787	0.732	0.7
LGB_CTriad	0.751	0.405	0.7	0.579	0.812	0.729	0.643	0.795	0.475	0.736	0.581	0.87	0.794	0.671
LGB_DDE	0.797	0.462	0.73	0.69	0.769	0.722	0.704	0.852	0.514	0.759	0.704	0.807	0.759	0.73
LGB_DistancePair	0.802	0.463	0.73	0.67	0.787	0.733	0.699	0.827	0.483	0.744	0.704	0.778	0.733	0.718
LGB_DPC-type-1	0.777	0.43	0.72	0.632	0.79	0.724	0.673	0.829	0.52	0.762	0.687	0.826	0.774	0.728
LGB_DPC-type-2	0.781	0.426	0.71	0.583	0.826	0.745	0.652	0.817	0.456	0.728	0.587	0.85	0.772	0.667
LGB_GDPC-type-2	0.756	0.387	0.7	0.617	0.765	0.694	0.652	0.796	0.43	0.715	0.564	0.845	0.759	0.647
LGB_PAAC	0.805	0.496	0.75	0.683	0.807	0.755	0.715	0.843	0.51	0.757	0.682	0.821	0.767	0.722
LGB_PDE	0.836	0.542	0.77	0.73	0.808	0.767	0.746	0.885	0.588	0.795	0.76	0.826	0.791	0.775
LGB_QSOrder	0.812	0.476	0.74	0.683	0.789	0.736	0.707	0.858	0.567	0.785	0.743	0.821	0.782	0.762
LRT_Bepler	0.769	0.409	0.71	0.659	0.748	0.692	0.674	0.74	0.351	0.679	0.587	0.759	0.677	0.629

LRT_CPCProt	0.765	0.449	0.73	0.685	0.761	0.712	0.698	0.799	0.499	0.751	0.698	0.797	0.749	0.723
LRT_ESM	0.837	0.536	0.77	0.737	0.796	0.758	0.746	0.872	0.625	0.814	0.754	0.865	0.828	0.79
LRT_ESM1b	0.831	0.522	0.76	0.73	0.789	0.75	0.738	0.849	0.609	0.806	0.782	0.826	0.796	0.789
LRT_ESM1v	0.831	0.54	0.77	0.74	0.797	0.759	0.749	0.846	0.567	0.785	0.721	0.841	0.796	0.757
LRT_ESM2-t6	0.825	0.517	0.76	0.732	0.784	0.746	0.738	0.842	0.583	0.793	0.732	0.845	0.804	0.766
LRT_ESM2-t12	0.834	0.542	0.77	0.739	0.8	0.762	0.748	0.842	0.541	0.772	0.704	0.831	0.783	0.741
LRT_ESM2-t30	0.822	0.529	0.76	0.729	0.796	0.757	0.74	0.85	0.6	0.801	0.715	0.874	0.831	0.769
LRT_ESM2-t33	0.84	0.552	0.78	0.75	0.801	0.765	0.756	0.863	0.614	0.808	0.782	0.831	0.8	0.791
LRT_ESM2-t36	0.843	0.553	0.78	0.744	0.806	0.768	0.755	0.868	0.61	0.806	0.726	0.874	0.833	0.776
LRT_FastText	0.816	0.495	0.75	0.701	0.79	0.745	0.721	0.852	0.525	0.764	0.698	0.821	0.772	0.733
LRT_GloVe	0.791	0.452	0.73	0.671	0.777	0.723	0.694	0.794	0.442	0.723	0.682	0.759	0.709	0.695
LRT_PRNN	0.803	0.484	0.74	0.723	0.76	0.723	0.722	0.869	0.598	0.801	0.749	0.845	0.807	0.777
LRT_PTAB	0.834	0.526	0.76	0.737	0.786	0.75	0.743	0.815	0.51	0.757	0.682	0.821	0.767	0.722
LRT_PTBB	0.794	0.477	0.74	0.728	0.749	0.715	0.72	0.783	0.432	0.718	0.687	0.744	0.699	0.693
LRT_PTB	0.835	0.549	0.78	0.758	0.79	0.758	0.757	0.858	0.551	0.777	0.721	0.826	0.782	0.75
LRT_PTU	0.843	0.558	0.78	0.77	0.788	0.759	0.763	0.849	0.546	0.775	0.737	0.807	0.767	0.752
LRT_PTXU	0.845	0.554	0.78	0.744	0.806	0.771	0.755	0.862	0.557	0.78	0.721	0.831	0.787	0.752
LRT_PTXLU	0.785	0.434	0.72	0.694	0.739	0.697	0.695	0.82	0.514	0.759	0.71	0.802	0.756	0.732
LRT_Seq2Vec	0.855	0.567	0.78	0.744	0.819	0.782	0.762	0.877	0.615	0.808	0.737	0.87	0.83	0.781
LRT_Word2Vec	0.803	0.473	0.74	0.692	0.777	0.729	0.709	0.834	0.525	0.764	0.71	0.812	0.765	0.736
LRT_AAC	0.683	0.287	0.65	0.507	0.768	0.655	0.57	0.668	0.26	0.635	0.508	0.744	0.632	0.564
LRT_APAAC	0.772	0.427	0.71	0.652	0.769	0.712	0.678	0.764	0.42	0.712	0.62	0.792	0.721	0.667
LRT_ASDC	0.739	0.378	0.69	0.641	0.734	0.677	0.657	0.796	0.467	0.736	0.659	0.802	0.742	0.698
LRT_CKSAAGP-type-1	0.704	0.314	0.66	0.59	0.721	0.645	0.615	0.723	0.33	0.668	0.587	0.739	0.66	0.621
LRT_CKSAAGP-type-2	0.73	0.361	0.68	0.556	0.792	0.698	0.618	0.762	0.354	0.679	0.514	0.821	0.713	0.597
LRT_CKSAAP-type-1	0.782	0.411	0.71	0.667	0.74	0.691	0.677	0.783	0.472	0.738	0.682	0.787	0.735	0.707
LRT_CKSAAP-type-2	0.822	0.501	0.75	0.712	0.785	0.743	0.726	0.815	0.531	0.767	0.682	0.841	0.787	0.731
LRT_CTDC	0.684	0.294	0.65	0.51	0.772	0.66	0.574	0.674	0.299	0.653	0.497	0.787	0.669	0.571
LRT_CTDD	0.725	0.37	0.69	0.635	0.733	0.672	0.652	0.744	0.399	0.702	0.62	0.773	0.703	0.659
LRT_CTDT	0.675	0.273	0.64	0.497	0.764	0.646	0.561	0.67	0.228	0.619	0.475	0.744	0.616	0.536
LRT_CTriad	0.704	0.352	0.68	0.565	0.777	0.688	0.618	0.71	0.353	0.679	0.536	0.802	0.701	0.608
LRT_DDE	0.785	0.441	0.72	0.67	0.768	0.713	0.691	0.802	0.463	0.733	0.626	0.826	0.757	0.685
LRT_DistancePair	0.683	0.287	0.65	0.507	0.768	0.655	0.57	0.668	0.26	0.635	0.508	0.744	0.632	0.564
LRT_DPC-type-1	0.738	0.374	0.69	0.642	0.73	0.672	0.656	0.75	0.357	0.681	0.609	0.744	0.673	0.639
LRT_DPC-type-2	0.763	0.43	0.72	0.674	0.754	0.702	0.687	0.805	0.489	0.746	0.665	0.816	0.758	0.708
LRT_GDPC-type-2	0.711	0.313	0.66	0.559	0.747	0.656	0.602	0.75	0.336	0.671	0.542	0.783	0.683	0.604
LRT_PAAC	0.777	0.42	0.71	0.629	0.783	0.716	0.667	0.766	0.426	0.715	0.609	0.807	0.732	0.665
LRT_PDE	0.781	0.447	0.73	0.663	0.779	0.722	0.69	0.778	0.441	0.723	0.648	0.787	0.725	0.684
LRT_QSOrder	0.789	0.463	0.73	0.671	0.786	0.732	0.698	0.803	0.467	0.736	0.654	0.807	0.745	0.696
RF_Bepler	0.795	0.47	0.74	0.678	0.786	0.735	0.704	0.818	0.469	0.736	0.631	0.826	0.758	0.689
RF_CPCProt	0.796	0.501	0.75	0.674	0.819	0.764	0.715	0.852	0.543	0.772	0.67	0.86	0.805	0.732
RF_ESM	0.843	0.55	0.78	0.694	0.845	0.796	0.74	0.851	0.558	0.78	0.693	0.855	0.805	0.745
RF_ESM1b	0.84	0.552	0.78	0.685	0.854	0.804	0.738	0.854	0.547	0.775	0.687	0.85	0.799	0.739
RF_ESM1v	0.84	0.552	0.78	0.706	0.838	0.791	0.746	0.835	0.522	0.762	0.654	0.855	0.796	0.718
RF_ESM2-t6	0.851	0.569	0.79	0.715	0.845	0.802	0.754	0.849	0.49	0.746	0.643	0.836	0.772	0.701
RF_ESM2-t12	0.839	0.556	0.78	0.711	0.838	0.792	0.748	0.85	0.547	0.775	0.698	0.841	0.791	0.742
RF_ESM2-t30	0.84	0.547	0.77	0.709	0.831	0.785	0.743	0.836	0.495	0.749	0.648	0.836	0.773	0.705
RF_ESM2-t33	0.854	0.563	0.78	0.718	0.838	0.793	0.752	0.868	0.637	0.819	0.743	0.884	0.847	0.792
RF_ESM2-t36	0.837	0.544	0.77	0.684	0.848	0.798	0.735	0.863	0.586	0.793	0.693	0.879	0.832	0.756
RF_FastText	0.822	0.53	0.77	0.673	0.845	0.791	0.726	0.829	0.537	0.769	0.682	0.845	0.792	0.733
RF_GloVe	0.793	0.47	0.74	0.642	0.818	0.752	0.691	0.803	0.442	0.723	0.609	0.821	0.747	0.671
RF_PRNN	0.824	0.519	0.76	0.66	0.845	0.788	0.717	0.886	0.561	0.78	0.665	0.879	0.826	0.737
RF_PTAB	0.835	0.545	0.77	0.698	0.837	0.789	0.739	0.822	0.485	0.744	0.637	0.836	0.77	0.697
RF_PTBB	0.804	0.501	0.75	0.669	0.824	0.767	0.713	0.808	0.451	0.728	0.654	0.792	0.731	0.69
RF_PTB	0.842	0.562	0.78	0.721	0.835	0.792	0.753	0.844	0.547	0.775	0.693	0.845	0.795	0.74
RF_PTU	0.852	0.577	0.79	0.719	0.849	0.805	0.758	0.853	0.588	0.795	0.737	0.845	0.805	0.77
RF_PTXU	0.853	0.571	0.79	0.726	0.838	0.797	0.758	0.853	0.551	0.777	0.715	0.831	0.785	0.749

RF_PTXLU	0.779	0.443	0.72	0.631	0.803	0.735	0.677	0.813	0.458	0.731	0.62	0.826	0.755	0.681
RF_Seq2Vec	0.854	0.564	0.78	0.704	0.85	0.803	0.748	0.876	0.567	0.782	0.665	0.884	0.832	0.739
RF_Word2Vec	0.808	0.492	0.75	0.659	0.824	0.764	0.707	0.814	0.495	0.749	0.654	0.831	0.77	0.707
RF_AAC	0.806	0.48	0.74	0.669	0.804	0.749	0.705	0.835	0.499	0.751	0.698	0.797	0.749	0.723
RF_APAAC	0.815	0.505	0.75	0.684	0.814	0.762	0.719	0.848	0.542	0.772	0.687	0.845	0.794	0.737
RF_ASDC	0.82	0.512	0.76	0.702	0.804	0.757	0.728	0.864	0.536	0.769	0.693	0.836	0.785	0.736
RF_CKSAAGP-type-1	0.799	0.494	0.75	0.697	0.792	0.745	0.718	0.843	0.527	0.764	0.67	0.845	0.79	0.725
RF_CKSAAGP-type-2	0.809	0.49	0.75	0.701	0.785	0.739	0.718	0.852	0.531	0.767	0.693	0.831	0.78	0.734
RF_CKSAAP-type-1	0.82	0.526	0.76	0.691	0.826	0.777	0.729	0.846	0.499	0.751	0.665	0.826	0.768	0.713
RF_CKSAAP-type-2	0.832	0.526	0.76	0.666	0.848	0.79	0.722	0.849	0.507	0.754	0.643	0.85	0.788	0.708
RF_CTDC	0.808	0.474	0.74	0.68	0.79	0.736	0.706	0.819	0.505	0.754	0.67	0.826	0.769	0.716
RF_CTDD	0.823	0.514	0.76	0.697	0.812	0.762	0.726	0.851	0.553	0.777	0.687	0.855	0.804	0.741
RF_CTDT	0.786	0.457	0.73	0.663	0.789	0.732	0.694	0.798	0.451	0.728	0.654	0.792	0.731	0.69
RF_CTriad	0.779	0.443	0.72	0.643	0.792	0.729	0.682	0.822	0.517	0.759	0.648	0.855	0.795	0.714
RF_DDE	0.814	0.506	0.75	0.705	0.797	0.75	0.726	0.863	0.551	0.777	0.726	0.821	0.778	0.751
RF_DistancePair	0.806	0.48	0.74	0.669	0.804	0.749	0.705	0.835	0.499	0.751	0.698	0.797	0.749	0.723
RF_DPC-type-1	0.812	0.51	0.76	0.695	0.809	0.759	0.725	0.854	0.578	0.79	0.726	0.845	0.803	0.763
RF_DPC-type-2	0.822	0.516	0.76	0.667	0.838	0.78	0.718	0.832	0.481	0.741	0.62	0.845	0.776	0.689
RF_GDPC-type-2	0.778	0.421	0.71	0.66	0.757	0.7	0.678	0.832	0.499	0.751	0.682	0.812	0.758	0.718
RF_PAAC	0.814	0.504	0.75	0.694	0.804	0.756	0.722	0.835	0.457	0.731	0.659	0.792	0.733	0.694
RF_PDE	0.842	0.56	0.78	0.743	0.814	0.775	0.758	0.882	0.593	0.798	0.777	0.816	0.785	0.781
RF_QSOrder	0.826	0.494	0.75	0.671	0.815	0.759	0.711	0.857	0.551	0.777	0.71	0.836	0.789	0.747
XGB_Bepler	0.801	0.492	0.75	0.711	0.779	0.736	0.722	0.84	0.521	0.762	0.67	0.841	0.784	0.723
XGB_CPCProt	0.802	0.502	0.75	0.701	0.797	0.749	0.723	0.849	0.546	0.775	0.726	0.816	0.774	0.749
XGB_ESM	0.848	0.563	0.78	0.74	0.819	0.78	0.758	0.861	0.583	0.793	0.732	0.845	0.804	0.766
XGB_ESM1b	0.858	0.56	0.78	0.736	0.82	0.779	0.755	0.872	0.556	0.78	0.732	0.821	0.78	0.755
XGB_ESM1v	0.841	0.559	0.78	0.747	0.809	0.772	0.759	0.858	0.547	0.775	0.687	0.85	0.799	0.739
XGB_ESM2-t6	0.869	0.6	0.8	0.767	0.83	0.795	0.779	0.858	0.567	0.785	0.749	0.816	0.779	0.764
XGB_ESM2-t12	0.839	0.557	0.78	0.739	0.814	0.774	0.754	0.845	0.562	0.782	0.726	0.831	0.788	0.756
XGB_ESM2-t30	0.845	0.567	0.78	0.742	0.821	0.782	0.76	0.855	0.535	0.769	0.71	0.821	0.774	0.741
XGB_ESM2-t33	0.866	0.58	0.79	0.743	0.832	0.794	0.766	0.884	0.635	0.819	0.76	0.87	0.834	0.795
XGB_ESM2-t36	0.845	0.546	0.77	0.718	0.821	0.778	0.744	0.886	0.622	0.811	0.726	0.884	0.844	0.781
XGB_FastText	0.831	0.527	0.76	0.713	0.809	0.765	0.737	0.843	0.535	0.769	0.71	0.821	0.774	0.741
XGB_GloVe	0.822	0.508	0.76	0.705	0.8	0.752	0.727	0.819	0.447	0.725	0.682	0.763	0.714	0.697
XGB_PRNN	0.829	0.547	0.77	0.734	0.809	0.769	0.751	0.879	0.614	0.808	0.749	0.86	0.822	0.784
XGB_PTAB	0.846	0.569	0.79	0.736	0.827	0.788	0.76	0.83	0.536	0.769	0.698	0.831	0.781	0.738
XGB_PTBB	0.801	0.49	0.75	0.707	0.78	0.737	0.72	0.767	0.441	0.723	0.67	0.768	0.714	0.692
XGB_PTB	0.847	0.564	0.78	0.746	0.815	0.778	0.76	0.856	0.572	0.788	0.749	0.821	0.784	0.766
XGB_PTU	0.869	0.579	0.79	0.747	0.827	0.789	0.766	0.854	0.572	0.788	0.732	0.836	0.794	0.762
XGB_PTXU	0.855	0.576	0.79	0.763	0.812	0.777	0.769	0.866	0.546	0.775	0.715	0.826	0.781	0.746
XGB_PTXLU	0.805	0.487	0.74	0.709	0.775	0.732	0.72	0.807	0.499	0.751	0.698	0.797	0.749	0.723
XGB_Seq2Vec	0.866	0.598	0.8	0.773	0.824	0.791	0.781	0.894	0.652	0.826	0.754	0.889	0.854	0.801
XGB_Word2Vec	0.816	0.512	0.76	0.728	0.783	0.744	0.734	0.849	0.541	0.772	0.726	0.812	0.769	0.747
XGB_AAC	0.805	0.476	0.74	0.69	0.783	0.733	0.71	0.831	0.536	0.769	0.732	0.802	0.762	0.746
XGB_APAAC	0.816	0.495	0.75	0.708	0.784	0.74	0.722	0.826	0.493	0.749	0.693	0.797	0.747	0.719
XGB_ASDC	0.81	0.501	0.75	0.709	0.789	0.744	0.725	0.849	0.562	0.782	0.737	0.821	0.781	0.759
XGB_CKSAAGP-type-1	0.788	0.466	0.73	0.687	0.774	0.728	0.704	0.851	0.504	0.754	0.676	0.821	0.766	0.718
XGB_CKSAAGP-type-2	0.796	0.465	0.73	0.68	0.781	0.729	0.703	0.817	0.484	0.744	0.643	0.831	0.767	0.699
XGB_CKSAAP-type-1	0.808	0.491	0.75	0.677	0.807	0.754	0.711	0.825	0.484	0.744	0.643	0.831	0.767	0.699
XGB_CKSAAP-type-2	0.825	0.523	0.76	0.701	0.815	0.769	0.731	0.832	0.504	0.754	0.693	0.807	0.756	0.723
XGB_CTDC	0.797	0.465	0.73	0.687	0.774	0.726	0.704	0.812	0.472	0.738	0.676	0.792	0.738	0.706
XGB_CTDD	0.816	0.503	0.75	0.698	0.801	0.751	0.722	0.842	0.499	0.751	0.698	0.797	0.749	0.723
XGB_CTDT	0.784	0.451	0.73	0.687	0.762	0.715	0.7	0.794	0.436	0.72	0.67	0.763	0.71	0.69
XGB_CTriad	0.763	0.434	0.72	0.649	0.78	0.718	0.681	0.808	0.495	0.749	0.643	0.841	0.777	0.703
XGB_DDE	0.821	0.51	0.76	0.709	0.797	0.751	0.729	0.861	0.583	0.793	0.771	0.812	0.78	0.775
XGB_DistancePair	0.805	0.476	0.74	0.69	0.783	0.733	0.71	0.831	0.536	0.769	0.732	0.802	0.762	0.746
XGB_DPC-type-1	0.81	0.489	0.75	0.698	0.787	0.74	0.717	0.838	0.536	0.769	0.749	0.787	0.753	0.751

XGB_DPC-type-2	0.808	0.492	0.75	0.699	0.787	0.742	0.718	0.822	0.499	0.751	0.67	0.821	0.764	0.714
XGB_GDPC-type-2	0.771	0.432	0.72	0.659	0.769	0.712	0.683	0.815	0.441	0.723	0.631	0.802	0.734	0.679
XGB_PAAC	0.817	0.508	0.76	0.711	0.793	0.749	0.728	0.835	0.483	0.744	0.676	0.802	0.747	0.71
XGB_PDE	0.845	0.56	0.78	0.746	0.812	0.773	0.759	0.864	0.61	0.806	0.793	0.816	0.789	0.791
XGB_QSOrder	0.827	0.519	0.76	0.722	0.793	0.752	0.735	0.858	0.583	0.793	0.749	0.831	0.793	0.77

TABLE S8

PERFORMANCE OF 360 BASELINE SINGLE FEATURE-MODELS ON THE TRAINING DATASET AND INDEPENDENT DATASET FOR THE REGRESSION TASK

Regression models	Training					Independent				
	MAE	RMSE	MSE	R	R^2	MAE	RMSE	MSE	R	R^2
AB_AAC	0.492	0.629	0.396	0.326	0.087	0.495	0.61	0.373	0.323	0.081
AB_APAAC	0.467	0.591	0.35	0.472	0.193	0.486	0.589	0.347	0.416	0.143
AB_ASDC	0.477	0.62	0.385	0.382	0.114	0.475	0.587	0.344	0.446	0.151
AB_Bepler	0.47	0.606	0.367	0.419	0.154	0.471	0.583	0.34	0.425	0.161
AB_CKSAAGP-type-1	0.468	0.601	0.362	0.443	0.166	0.471	0.583	0.34	0.46	0.161
AB_CKSAAGP-type-2	0.467	0.597	0.357	0.443	0.177	0.467	0.577	0.333	0.451	0.179
AB_CKSAAP-type-1	0.478	0.628	0.394	0.327	0.092	0.494	0.616	0.379	0.311	0.064
AB_CKSAAP-type-2	0.488	0.631	0.399	0.33	0.08	0.502	0.615	0.378	0.284	0.068
AB_CPCProt	0.459	0.6	0.36	0.454	0.171	0.465	0.58	0.336	0.443	0.171
AB_CTDC	0.467	0.606	0.367	0.409	0.155	0.486	0.601	0.361	0.342	0.108
AB_CTDD	0.452	0.584	0.341	0.492	0.212	0.452	0.555	0.308	0.538	0.241
AB_CTDT	0.48	0.614	0.378	0.399	0.129	0.501	0.612	0.374	0.345	0.077
AB_CTriad	0.485	0.62	0.385	0.344	0.112	0.512	0.626	0.392	0.22	0.033
AB_DDE	0.475	0.61	0.372	0.399	0.143	0.48	0.585	0.343	0.423	0.155
AB_DPC-type-1	0.494	0.632	0.4	0.345	0.078	0.513	0.624	0.389	0.243	0.04
AB_DPC-type-2	0.495	0.638	0.407	0.31	0.062	0.509	0.627	0.393	0.196	0.031
AB_DistancePair	0.492	0.629	0.396	0.326	0.087	0.495	0.61	0.373	0.323	0.081
AB_ESM	0.425	0.554	0.307	0.574	0.29	0.438	0.549	0.301	0.528	0.257
AB_ESM1b	0.432	0.562	0.316	0.553	0.27	0.444	0.554	0.307	0.516	0.243
AB_ESM1v	0.435	0.57	0.325	0.543	0.251	0.444	0.558	0.312	0.51	0.231
AB_ESM2-t12	0.456	0.6	0.36	0.456	0.169	0.457	0.573	0.328	0.475	0.191
AB_ESM2-t30	0.442	0.574	0.33	0.521	0.239	0.447	0.559	0.312	0.501	0.23
AB_ESM2-t33	0.422	0.55	0.302	0.587	0.302	0.427	0.532	0.283	0.575	0.303
AB_ESM2-t36	0.431	0.559	0.313	0.563	0.278	0.431	0.542	0.294	0.558	0.274
AB_ESM2-t6	0.436	0.568	0.322	0.542	0.256	0.449	0.563	0.317	0.486	0.219
AB_FastText	0.444	0.577	0.333	0.526	0.232	0.449	0.564	0.318	0.509	0.215
AB_GDPC-type-2	0.488	0.62	0.385	0.353	0.11	0.491	0.597	0.357	0.367	0.12
AB_GloVe	0.458	0.597	0.357	0.457	0.177	0.457	0.569	0.323	0.496	0.202
AB_PAAC	0.465	0.596	0.355	0.46	0.18	0.475	0.579	0.335	0.465	0.172
AB_PDE	0.47	0.588	0.346	0.506	0.2	0.491	0.591	0.35	0.49	0.137
AB_PRNN	0.434	0.568	0.322	0.536	0.256	0.443	0.551	0.304	0.54	0.25
AB_PTAB	0.433	0.567	0.321	0.55	0.259	0.45	0.559	0.313	0.497	0.228
AB_PTB	0.43	0.561	0.314	0.564	0.275	0.437	0.545	0.297	0.552	0.268
AB_PTBB	0.464	0.608	0.369	0.441	0.148	0.476	0.595	0.354	0.401	0.126
AB_PTU	0.424	0.555	0.308	0.564	0.289	0.426	0.532	0.283	0.572	0.301
AB_PTXLU	0.462	0.599	0.359	0.443	0.172	0.476	0.592	0.351	0.398	0.135
AB_PTXU	0.425	0.554	0.307	0.575	0.292	0.436	0.542	0.294	0.546	0.274
AB_QSOrder	0.476	0.601	0.361	0.476	0.166	0.496	0.594	0.353	0.457	0.128
AB_Seq2Vec	0.427	0.553	0.306	0.57	0.294	0.426	0.537	0.288	0.567	0.289
AB_Word2Vec	0.451	0.59	0.348	0.477	0.198	0.463	0.578	0.335	0.445	0.175
CB_AAC	0.385	0.54	0.291	0.585	0.327	0.37	0.507	0.257	0.628	0.366
CB_APAAC	0.375	0.516	0.266	0.625	0.383	0.355	0.468	0.219	0.68	0.46
CB_ASDC	0.391	0.527	0.278	0.601	0.359	0.353	0.472	0.222	0.673	0.451
CB_Bepler	0.402	0.541	0.293	0.574	0.324	0.388	0.491	0.241	0.637	0.405
CB_CKSAAGP-type-1	0.403	0.546	0.298	0.564	0.313	0.373	0.49	0.24	0.64	0.408
CB_CKSAAGP-type-2	0.415	0.554	0.307	0.545	0.291	0.387	0.496	0.246	0.638	0.394
CB_CKSAAP-type-1	0.389	0.532	0.283	0.592	0.348	0.372	0.489	0.239	0.641	0.41

CB_CKSAAP-type-2	0.377	0.522	0.273	0.616	0.371	0.339	0.45	0.203	0.71	0.499
CB_CPCProt	0.409	0.548	0.301	0.557	0.304	0.382	0.507	0.257	0.606	0.366
CB_CTDC	0.392	0.537	0.288	0.582	0.335	0.37	0.507	0.257	0.614	0.365
CB_CTDD	0.391	0.528	0.279	0.6	0.354	0.368	0.483	0.233	0.652	0.424
CB_CTDT	0.405	0.554	0.306	0.547	0.291	0.383	0.514	0.264	0.598	0.348
CB_CTriad	0.42	0.566	0.32	0.517	0.26	0.41	0.524	0.275	0.57	0.322
CB_DDE	0.384	0.522	0.272	0.617	0.37	0.362	0.473	0.223	0.671	0.449
CB_DPC-type-1	0.391	0.535	0.286	0.59	0.34	0.369	0.487	0.237	0.645	0.415
CB_DPC-type-2	0.381	0.527	0.277	0.618	0.358	0.355	0.479	0.229	0.667	0.434
CB_DistancePair	0.398	0.546	0.299	0.567	0.311	0.373	0.502	0.252	0.624	0.377
CB_ESM	0.374	0.51	0.26	0.633	0.397	0.366	0.473	0.224	0.669	0.447
CB_ESM1b	0.37	0.492	0.242	0.665	0.441	0.362	0.474	0.225	0.668	0.446
CB_ESM1v	0.375	0.507	0.257	0.638	0.403	0.369	0.486	0.236	0.647	0.418
CB_ESM2-t12	0.369	0.497	0.247	0.657	0.427	0.358	0.47	0.221	0.677	0.454
CB_ESM2-t30	0.374	0.51	0.26	0.638	0.396	0.394	0.513	0.264	0.606	0.35
CB_ESM2-t33	0.361	0.487	0.237	0.673	0.45	0.339	0.447	0.2	0.715	0.508
CB_ESM2-t36	0.37	0.5	0.25	0.65	0.422	0.357	0.458	0.21	0.695	0.482
CB_ESM2-t6	0.369	0.499	0.249	0.653	0.421	0.375	0.489	0.239	0.643	0.411
CB_FastText	0.384	0.529	0.28	0.602	0.353	0.37	0.485	0.236	0.648	0.418
CB_GDPC-type-2	0.437	0.579	0.335	0.481	0.225	0.416	0.522	0.273	0.582	0.327
CB_GloVe	0.396	0.544	0.295	0.574	0.317	0.373	0.488	0.238	0.644	0.413
CB_PAAC	0.383	0.53	0.281	0.602	0.349	0.365	0.475	0.225	0.667	0.444
CB_PDE	0.391	0.546	0.298	0.598	0.311	0.368	0.485	0.236	0.659	0.419
CB_PRNN	0.385	0.52	0.27	0.614	0.374	0.352	0.466	0.217	0.683	0.464
CB_PTAB	0.373	0.5	0.25	0.649	0.42	0.375	0.485	0.235	0.65	0.42
CB_PTB	0.375	0.509	0.259	0.636	0.399	0.367	0.474	0.225	0.668	0.445
CB_PTBB	0.404	0.543	0.295	0.569	0.316	0.405	0.535	0.286	0.552	0.295
CB_PTU	0.353	0.483	0.233	0.683	0.461	0.368	0.485	0.235	0.653	0.419
CB_PTXLU	0.415	0.547	0.299	0.558	0.309	0.407	0.513	0.263	0.594	0.351
CB_PTXU	0.364	0.493	0.243	0.662	0.437	0.366	0.476	0.226	0.665	0.441
CB_QSOrder	0.382	0.523	0.273	0.616	0.368	0.376	0.485	0.235	0.651	0.421
CB_Seq2Vec	0.346	0.47	0.221	0.699	0.487	0.34	0.449	0.201	0.71	0.503
CB_Word2Vec	0.401	0.551	0.303	0.557	0.298	0.38	0.503	0.253	0.618	0.376
ERT_AAC	0.383	0.542	0.294	0.575	0.323	0.382	0.503	0.253	0.624	0.376
ERT_APAAC	0.389	0.542	0.293	0.58	0.323	0.374	0.485	0.235	0.675	0.419
ERT_ASDC	0.38	0.52	0.27	0.627	0.375	0.38	0.49	0.24	0.671	0.407
ERT_Bepler	0.434	0.571	0.326	0.572	0.249	0.428	0.534	0.286	0.623	0.296
ERT_CKSAAGP-type-1	0.394	0.542	0.294	0.574	0.322	0.371	0.475	0.226	0.689	0.443
ERT_CKSAAGP-type-2	0.395	0.551	0.304	0.552	0.3	0.37	0.473	0.224	0.692	0.447
ERT_CKSAAP-type-1	0.388	0.53	0.28	0.606	0.353	0.371	0.479	0.23	0.693	0.434
ERT_CKSAAP-type-2	0.379	0.528	0.279	0.608	0.355	0.369	0.486	0.236	0.669	0.417
ERT_CPCProt	0.416	0.575	0.33	0.499	0.236	0.39	0.512	0.262	0.613	0.354
ERT_CTDC	0.389	0.551	0.303	0.55	0.301	0.368	0.499	0.249	0.622	0.386
ERT_CTDD	0.392	0.53	0.281	0.617	0.351	0.372	0.478	0.229	0.711	0.435
ERT_CTDT	0.406	0.564	0.319	0.518	0.265	0.397	0.532	0.284	0.549	0.3
ERT_CTriad	0.41	0.571	0.326	0.505	0.248	0.389	0.508	0.258	0.618	0.362
ERT_DDE	0.389	0.533	0.284	0.59	0.344	0.368	0.47	0.221	0.69	0.456
ERT_DPC-type-1	0.387	0.532	0.283	0.593	0.346	0.369	0.483	0.233	0.671	0.425
ERT_DPC-type-2	0.379	0.527	0.278	0.607	0.358	0.37	0.495	0.245	0.645	0.396
ERT_DistancePair	0.383	0.542	0.294	0.575	0.323	0.382	0.503	0.253	0.624	0.376
ERT_ESM	0.408	0.546	0.299	0.613	0.31	0.414	0.524	0.275	0.632	0.321
ERT_ESM1b	0.402	0.534	0.285	0.639	0.341	0.409	0.523	0.273	0.634	0.325
ERT_ESM1v	0.412	0.549	0.302	0.622	0.304	0.416	0.527	0.278	0.645	0.315
ERT_ESM2-t12	0.412	0.548	0.3	0.617	0.306	0.417	0.526	0.277	0.634	0.317
ERT_ESM2-t30	0.404	0.538	0.29	0.623	0.331	0.41	0.519	0.269	0.627	0.336
ERT_ESM2-t33	0.397	0.531	0.281	0.65	0.35	0.404	0.51	0.26	0.655	0.358
ERT_ESM2-t36	0.403	0.534	0.285	0.645	0.342	0.41	0.518	0.269	0.653	0.337

ERT_ESM2-t6	0.4	0.536	0.287	0.638	0.337	0.411	0.52	0.27	0.638	0.333
ERT_FastText	0.415	0.565	0.319	0.545	0.264	0.412	0.52	0.27	0.624	0.333
ERT_GDPC-type-2	0.404	0.563	0.317	0.524	0.269	0.372	0.485	0.235	0.654	0.42
ERT_GloVe	0.424	0.576	0.332	0.507	0.234	0.416	0.528	0.279	0.596	0.312
ERT_PAAC	0.378	0.531	0.281	0.602	0.35	0.376	0.493	0.243	0.654	0.4
ERT_PDE	0.367	0.513	0.264	0.64	0.392	0.354	0.462	0.214	0.71	0.473
ERT_PRNN	0.419	0.56	0.313	0.603	0.278	0.412	0.517	0.268	0.69	0.339
ERT_PTAB	0.414	0.546	0.298	0.649	0.312	0.426	0.534	0.285	0.659	0.296
ERT_PTB	0.414	0.547	0.299	0.636	0.31	0.426	0.535	0.286	0.643	0.295
ERT_PTBB	0.439	0.58	0.337	0.563	0.223	0.451	0.566	0.32	0.573	0.211
ERT_PTU	0.409	0.541	0.293	0.641	0.325	0.413	0.526	0.277	0.638	0.316
ERT_PTXLU	0.439	0.573	0.328	0.566	0.243	0.445	0.555	0.308	0.604	0.24
ERT_PTXU	0.406	0.539	0.29	0.639	0.33	0.407	0.514	0.264	0.657	0.348
ERT_QSOrder	0.39	0.542	0.293	0.585	0.323	0.383	0.493	0.243	0.669	0.401
ERT_Seq2Vec	0.403	0.535	0.286	0.656	0.339	0.409	0.517	0.267	0.685	0.341
ERT_Word2Vec	0.422	0.572	0.327	0.524	0.246	0.419	0.529	0.28	0.608	0.31
GB_AAC	0.393	0.54	0.292	0.577	0.327	0.388	0.522	0.272	0.59	0.328
GB_APAAC	0.39	0.534	0.285	0.591	0.342	0.365	0.476	0.226	0.665	0.442
GB_ASDC	0.406	0.544	0.296	0.564	0.317	0.372	0.489	0.239	0.645	0.41
GB_Bepler	0.418	0.556	0.309	0.545	0.288	0.405	0.508	0.258	0.611	0.364
GB_CKSAAGP-type-1	0.403	0.545	0.297	0.564	0.314	0.366	0.483	0.233	0.652	0.424
GB_CKSAAGP-type-2	0.412	0.56	0.313	0.536	0.276	0.376	0.486	0.236	0.648	0.418
GB_CKSAAP-type-1	0.416	0.552	0.305	0.553	0.296	0.42	0.534	0.285	0.553	0.297
GB_CKSAAP-type-2	0.422	0.557	0.31	0.547	0.285	0.414	0.518	0.268	0.595	0.338
GB_CPCProt	0.421	0.559	0.313	0.535	0.278	0.406	0.528	0.279	0.564	0.312
GB_CTDC	0.387	0.539	0.291	0.577	0.327	0.374	0.507	0.257	0.613	0.366
GB_CTDD	0.399	0.536	0.288	0.579	0.335	0.368	0.48	0.23	0.662	0.432
GB_CTDT	0.404	0.553	0.305	0.548	0.294	0.389	0.529	0.28	0.57	0.308
GB_CTriad	0.427	0.574	0.33	0.491	0.239	0.422	0.537	0.289	0.537	0.287
GB_DDE	0.419	0.549	0.302	0.559	0.303	0.412	0.514	0.265	0.601	0.347
GB_DPC-type-1	0.404	0.539	0.291	0.576	0.329	0.386	0.504	0.254	0.613	0.373
GB_DPC-type-2	0.407	0.544	0.296	0.566	0.315	0.372	0.489	0.239	0.641	0.409
GB_DistancePair	0.393	0.54	0.292	0.577	0.327	0.388	0.522	0.272	0.59	0.328
GB_ESM	0.401	0.528	0.279	0.605	0.355	0.408	0.517	0.267	0.592	0.341
GB_ESM1b	0.41	0.535	0.286	0.593	0.339	0.402	0.51	0.26	0.615	0.359
GB_ESM1v	0.414	0.543	0.294	0.578	0.319	0.423	0.531	0.282	0.56	0.305
GB_ESM2-t12	0.415	0.541	0.293	0.582	0.323	0.423	0.524	0.275	0.585	0.322
GB_ESM2-t30	0.416	0.547	0.3	0.567	0.306	0.425	0.536	0.287	0.548	0.292
GB_ESM2-t33	0.394	0.518	0.268	0.628	0.379	0.39	0.494	0.244	0.641	0.398
GB_ESM2-t36	0.409	0.533	0.284	0.6	0.344	0.398	0.506	0.256	0.625	0.368
GB_ESM2-t6	0.401	0.524	0.275	0.612	0.362	0.412	0.524	0.275	0.573	0.323
GB_FastText	0.408	0.543	0.295	0.577	0.318	0.415	0.53	0.281	0.559	0.306
GB_GDPC-type-2	0.436	0.588	0.346	0.471	0.199	0.403	0.517	0.267	0.584	0.34
GB_GloVe	0.427	0.568	0.323	0.519	0.254	0.417	0.523	0.273	0.587	0.326
GB_PAAC	0.39	0.532	0.283	0.596	0.347	0.375	0.487	0.238	0.644	0.414
GB_PDE	0.363	0.506	0.256	0.643	0.408	0.35	0.466	0.217	0.683	0.464
GB_PRNN	0.412	0.542	0.294	0.575	0.32	0.412	0.521	0.272	0.586	0.33
GB_PTAB	0.414	0.543	0.295	0.58	0.318	0.418	0.523	0.273	0.59	0.326
GB_PTB	0.406	0.533	0.284	0.6	0.342	0.401	0.512	0.262	0.609	0.353
GB_PTBB	0.444	0.587	0.344	0.468	0.205	0.455	0.582	0.338	0.408	0.165
GB_PTU	0.393	0.515	0.266	0.639	0.386	0.398	0.503	0.253	0.624	0.376
GB_PTXLU	0.439	0.576	0.331	0.493	0.235	0.449	0.565	0.319	0.471	0.213
GB_PTXU	0.401	0.526	0.277	0.613	0.361	0.401	0.506	0.256	0.614	0.368
GB_QSOrder	0.387	0.534	0.285	0.591	0.342	0.383	0.495	0.245	0.632	0.395
GB_Seq2Vec	0.389	0.51	0.26	0.647	0.4	0.378	0.487	0.238	0.661	0.414
GB_Word2Vec	0.426	0.565	0.319	0.523	0.264	0.427	0.544	0.296	0.525	0.27
Lasso_AAC	0.477	0.631	0.399	0.29	0.081	0.484	0.62	0.384	0.248	0.052

Lasso_APAAC	0.442	0.571	0.326	0.503	0.247	0.453	0.566	0.32	0.462	0.211
Lasso_ASDC	0.447	0.593	0.352	0.441	0.188	0.427	0.571	0.326	0.449	0.195
Lasso_Bepler	0.451	0.585	0.342	0.474	0.21	0.446	0.565	0.32	0.471	0.211
Lasso_CKSAAGP-type-1	0.474	0.62	0.385	0.341	0.112	0.475	0.602	0.362	0.327	0.106
Lasso_CKSAAGP-type-2	0.468	0.6	0.361	0.414	0.168	0.457	0.567	0.321	0.456	0.207
Lasso_CKSAAP-type-1	0.433	0.581	0.338	0.481	0.221	0.418	0.564	0.318	0.49	0.215
Lasso_CKSAAP-type-2	0.403	0.537	0.288	0.585	0.335	0.364	0.473	0.224	0.669	0.447
Lasso_CPCProt	0.445	0.589	0.347	0.451	0.198	0.422	0.542	0.294	0.525	0.275
Lasso_CTDC	0.477	0.631	0.398	0.29	0.081	0.485	0.62	0.385	0.246	0.051
Lasso_CTDD	0.45	0.584	0.341	0.481	0.213	0.455	0.578	0.334	0.444	0.176
Lasso_CTDT	0.479	0.631	0.398	0.295	0.083	0.479	0.605	0.367	0.311	0.095
Lasso_CTriad	0.471	0.622	0.387	0.335	0.106	0.484	0.608	0.37	0.315	0.087
Lasso_DDE	0.437	0.568	0.323	0.51	0.255	0.765	3.116	9.707	0.043	-22.95
Lasso_DPC-type-1	0.461	0.605	0.366	0.405	0.155	0.464	0.594	0.352	0.388	0.13
Lasso_DPC-type-2	0.435	0.568	0.322	0.51	0.255	0.417	0.531	0.282	0.553	0.304
Lasso_DistancePair	0.477	0.631	0.399	0.29	0.081	0.484	0.62	0.384	0.248	0.052
Lasso_ESM	0.39	0.517	0.268	0.619	0.38	0.382	0.486	0.236	0.647	0.417
Lasso_ESM1b	0.403	0.529	0.28	0.596	0.351	0.409	0.525	0.275	0.568	0.321
Lasso_ESM1v	0.401	0.528	0.278	0.6	0.355	0.403	0.514	0.265	0.591	0.347
Lasso_ESM2-t12	0.397	0.524	0.275	0.623	0.363	0.379	0.491	0.241	0.646	0.406
Lasso_ESM2-t30	0.403	0.535	0.287	0.609	0.336	0.388	0.5	0.25	0.637	0.384
Lasso_ESM2-t33	0.403	0.532	0.283	0.592	0.346	0.398	0.503	0.253	0.613	0.375
Lasso_ESM2-t36	0.386	0.516	0.266	0.623	0.385	0.382	0.489	0.239	0.641	0.41
Lasso_ESM2-t6	0.398	0.534	0.285	0.604	0.338	0.392	0.51	0.26	0.608	0.359
Lasso_FastText	0.42	0.551	0.303	0.552	0.299	0.407	0.522	0.272	0.573	0.328
Lasso_GDPC-type-2	0.474	0.616	0.379	0.361	0.126	0.47	0.581	0.337	0.41	0.168
Lasso_GloVe	0.428	0.565	0.319	0.518	0.264	0.429	0.547	0.3	0.511	0.261
Lasso_PAAC	0.441	0.572	0.327	0.499	0.244	0.46	0.572	0.327	0.446	0.194
Lasso_PDE	0.44	0.573	0.328	0.499	0.241	0.453	0.565	0.319	0.467	0.213
Lasso_PRNN	0.403	0.532	0.283	0.591	0.345	0.39	0.494	0.244	0.631	0.397
Lasso_PTAB	0.402	0.526	0.277	0.599	0.357	0.397	0.5	0.25	0.62	0.383
Lasso_PTB	0.398	0.528	0.279	0.599	0.354	0.386	0.499	0.249	0.625	0.386
Lasso_PTBB	0.434	0.577	0.332	0.489	0.231	0.437	0.563	0.317	0.47	0.219
Lasso_PTU	0.389	0.517	0.268	0.622	0.381	0.393	0.505	0.255	0.61	0.371
Lasso_PTXLU	0.435	0.569	0.324	0.505	0.252	0.444	0.557	0.31	0.487	0.236
Lasso_PTXU	0.394	0.516	0.266	0.623	0.384	0.389	0.493	0.243	0.633	0.399
Lasso_QSOrder	0.441	0.574	0.329	0.496	0.239	0.44	0.551	0.304	0.501	0.25
Lasso_Seq2Vec	0.377	0.504	0.254	0.645	0.412	0.367	0.478	0.228	0.661	0.437
Lasso_Word2Vec	0.428	0.559	0.313	0.532	0.278	0.424	0.539	0.29	0.533	0.284
LGB_AAC	0.395	0.541	0.293	0.574	0.323	0.365	0.488	0.238	0.643	0.412
LGB_APAAC	0.378	0.515	0.265	0.625	0.387	0.352	0.464	0.215	0.686	0.468
LGB_ASDC	0.374	0.522	0.273	0.611	0.369	0.343	0.463	0.215	0.689	0.471
LGB_Bepler	0.392	0.532	0.283	0.594	0.347	0.364	0.477	0.227	0.663	0.44
LGB_CKSAAGP-type-1	0.398	0.547	0.3	0.556	0.308	0.364	0.484	0.234	0.65	0.422
LGB_CKSAAGP-type-2	0.397	0.545	0.297	0.564	0.315	0.361	0.477	0.228	0.663	0.438
LGB_CKSAAP-type-1	0.387	0.525	0.276	0.604	0.363	0.377	0.503	0.253	0.613	0.375
LGB_CKSAAP-type-2	0.368	0.506	0.256	0.641	0.408	0.321	0.442	0.195	0.721	0.519
LGB_CPCProt	0.394	0.537	0.288	0.582	0.333	0.353	0.483	0.234	0.651	0.423
LGB_CTDC	0.391	0.539	0.291	0.577	0.327	0.372	0.512	0.262	0.605	0.354
LGB_CTDD	0.365	0.506	0.256	0.638	0.405	0.348	0.467	0.218	0.682	0.463
LGB_CTDT	0.399	0.547	0.299	0.559	0.309	0.385	0.515	0.266	0.593	0.344
LGB_CTriad	0.422	0.564	0.318	0.522	0.266	0.398	0.517	0.268	0.585	0.34
LGB_DDE	0.373	0.513	0.264	0.629	0.391	0.348	0.457	0.208	0.698	0.486
LGB_DPC-type-1	0.395	0.539	0.29	0.578	0.331	0.366	0.489	0.239	0.64	0.41
LGB_DPC-type-2	0.395	0.535	0.286	0.587	0.339	0.354	0.474	0.224	0.669	0.447
LGB_DistancePair	0.395	0.541	0.293	0.574	0.323	0.365	0.488	0.238	0.643	0.412
LGB_ESM	0.371	0.502	0.252	0.646	0.415	0.369	0.478	0.228	0.662	0.437

LGB_ESM1b	0.372	0.498	0.248	0.654	0.426	0.349	0.459	0.211	0.696	0.48
LGB_ESM1v	0.372	0.499	0.249	0.656	0.424	0.368	0.483	0.233	0.653	0.425
LGB_ESM2-t12	0.361	0.489	0.239	0.667	0.444	0.356	0.462	0.213	0.69	0.473
LGB_ESM2-t30	0.37	0.507	0.257	0.638	0.404	0.365	0.475	0.226	0.668	0.443
LGB_ESM2-t33	0.35	0.479	0.229	0.688	0.468	0.336	0.439	0.192	0.728	0.525
LGB_ESM2-t36	0.377	0.505	0.255	0.645	0.41	0.361	0.464	0.215	0.691	0.468
LGB_ESM2-t6	0.358	0.491	0.241	0.665	0.44	0.361	0.477	0.228	0.663	0.438
LGB_FastText	0.377	0.52	0.271	0.619	0.374	0.367	0.482	0.232	0.655	0.427
LGB_GDPC-type-2	0.428	0.567	0.321	0.514	0.257	0.403	0.523	0.273	0.571	0.325
LGB_GloVe	0.383	0.532	0.283	0.597	0.346	0.36	0.47	0.221	0.675	0.454
LGB_PAAC	0.38	0.524	0.274	0.61	0.364	0.358	0.476	0.226	0.666	0.441
LGB_PDE	0.361	0.501	0.251	0.653	0.422	0.341	0.454	0.207	0.701	0.49
LGB_PRNN	0.373	0.506	0.256	0.64	0.406	0.344	0.452	0.204	0.707	0.496
LGB_PTAB	0.379	0.507	0.257	0.642	0.405	0.382	0.483	0.233	0.659	0.425
LGB_PTB	0.366	0.497	0.247	0.657	0.427	0.357	0.468	0.219	0.681	0.46
LGB_PTBB	0.403	0.544	0.296	0.568	0.316	0.402	0.521	0.272	0.577	0.33
LGB_PTU	0.357	0.479	0.23	0.688	0.468	0.365	0.476	0.226	0.666	0.442
LGB_PTXLU	0.407	0.542	0.294	0.566	0.32	0.395	0.512	0.262	0.596	0.354
LGB_PTXU	0.35	0.48	0.23	0.685	0.466	0.356	0.465	0.216	0.683	0.466
LGB_QSOrder	0.374	0.514	0.264	0.628	0.389	0.355	0.465	0.216	0.684	0.466
LGB_Seq2Vec	0.342	0.465	0.216	0.709	0.498	0.325	0.432	0.186	0.737	0.541
LGB_Word2Vec	0.395	0.539	0.29	0.578	0.329	0.381	0.497	0.247	0.626	0.391
ANN_AAC	0.426	0.58	0.336	0.499	0.225	0.436	0.573	0.329	0.508	0.189
ANN_APAAC	0.393	0.532	0.283	0.614	0.341	0.419	0.531	0.282	0.602	0.305
ANN_ASDC	0.467	0.652	0.425	0.464	0.011	0.425	0.599	0.359	0.544	0.115
ANN_Bepler	0.434	0.57	0.325	0.537	0.251	0.409	0.534	0.285	0.586	0.296
ANN_CKSAAGP-type-1	0.463	0.618	0.381	0.47	0.117	0.414	0.554	0.307	0.575	0.242
ANN_CKSAAGP-type-2	0.458	0.604	0.365	0.489	0.156	0.409	0.525	0.276	0.635	0.32
ANN_CKSAAP-type-1	0.526	0.716	0.513	0.433	-0.19	0.426	0.58	0.336	0.596	0.17
ANN_CKSAAP-type-2	0.497	0.679	0.461	0.5	-0.07	0.402	0.54	0.291	0.637	0.281
ANN_CPCProt	0.481	0.651	0.423	0.487	0.016	0.46	0.627	0.393	0.534	0.029
ANN_CTDC	0.435	0.585	0.343	0.502	0.209	0.431	0.562	0.316	0.533	0.221
ANN_CTDD	0.447	0.594	0.352	0.53	0.184	0.444	0.594	0.353	0.489	0.128
ANN_CTDT	0.444	0.587	0.345	0.487	0.201	0.461	0.596	0.355	0.469	0.124
ANN_CTriad	0.531	0.718	0.516	0.373	-0.19	0.506	0.659	0.434	0.451	-0.071
ANN_DDE	0.418	0.583	0.34	0.574	0.212	0.614	1.615	2.608	0.156	-5.435
ANN_DPC-type-1	0.476	0.653	0.427	0.448	0.012	0.474	0.659	0.435	0.438	-0.072
ANN_DPC-type-2	0.471	0.645	0.415	0.51	0.035	0.448	0.596	0.355	0.567	0.124
ANN_DistancePair	0.426	0.58	0.336	0.499	0.225	0.436	0.573	0.329	0.508	0.189
ANN_ESM	0.401	0.548	0.301	0.617	0.301	0.388	0.524	0.275	0.651	0.321
ANN_ESM1b	0.382	0.52	0.27	0.655	0.373	0.367	0.507	0.257	0.652	0.365
ANN_ESM1v	0.43	0.576	0.332	0.573	0.23	0.389	0.532	0.283	0.648	0.303
ANN_ESM2-t12	0.396	0.537	0.288	0.633	0.329	0.395	0.534	0.286	0.637	0.296
ANN_ESM2-t30	0.384	0.518	0.268	0.654	0.379	0.411	0.533	0.284	0.619	0.298
ANN_ESM2-t33	0.376	0.516	0.266	0.671	0.384	0.374	0.509	0.259	0.671	0.36
ANN_ESM2-t36	0.408	0.557	0.31	0.608	0.283	0.373	0.497	0.247	0.689	0.389
ANN_ESM2-t6	0.373	0.521	0.271	0.651	0.37	0.381	0.508	0.259	0.667	0.362
ANN_FastText	0.421	0.575	0.33	0.592	0.234	0.408	0.561	0.314	0.612	0.224
ANN_GDPC-type-2	0.45	0.59	0.349	0.454	0.193	0.441	0.557	0.31	0.494	0.235
ANN_GloVe	0.421	0.578	0.334	0.591	0.227	0.437	0.614	0.377	0.584	0.069
ANN_PAAC	0.398	0.539	0.29	0.605	0.327	0.4	0.538	0.289	0.599	0.286
ANN_PDE	0.393	0.534	0.285	0.605	0.341	0.38	0.517	0.267	0.649	0.342
ANN_PRNN	0.419	0.582	0.338	0.576	0.211	0.377	0.528	0.279	0.646	0.312
ANN_PTAB	0.423	0.571	0.327	0.582	0.241	0.423	0.559	0.312	0.605	0.229
ANN_PTB	0.411	0.556	0.309	0.627	0.285	0.395	0.528	0.279	0.625	0.312
ANN_PTBB	0.43	0.586	0.344	0.549	0.202	0.413	0.568	0.322	0.552	0.205
ANN_PTU	0.392	0.533	0.284	0.659	0.344	0.375	0.512	0.262	0.662	0.354

ANN_PTXLU	0.472	0.622	0.386	0.513	0.107	0.437	0.571	0.326	0.548	0.197
ANN_PTXU	0.402	0.546	0.298	0.636	0.31	0.392	0.523	0.273	0.659	0.326
ANN_QSOrder	0.395	0.538	0.289	0.613	0.331	0.398	0.534	0.285	0.618	0.296
ANN_Seq2Vec	0.371	0.506	0.256	0.688	0.409	0.391	0.535	0.286	0.667	0.294
ANN_Word2Vec	0.426	0.573	0.329	0.592	0.234	0.435	0.596	0.355	0.585	0.124
RF_AAC	0.387	0.537	0.289	0.585	0.334	0.37	0.491	0.241	0.643	0.405
RF_APAAC	0.396	0.538	0.29	0.59	0.331	0.373	0.484	0.234	0.671	0.422
RF_ASDC	0.385	0.521	0.272	0.629	0.373	0.362	0.475	0.226	0.687	0.443
RF_Bepler	0.437	0.574	0.329	0.542	0.241	0.427	0.534	0.285	0.598	0.297
RF_CKSAAGP-type-1	0.39	0.535	0.286	0.588	0.34	0.354	0.462	0.213	0.703	0.474
RF_CKSAAGP-type-2	0.396	0.544	0.296	0.573	0.318	0.372	0.478	0.228	0.682	0.436
RF_CKSAAP-type-1	0.39	0.529	0.28	0.613	0.355	0.363	0.471	0.222	0.701	0.452
RF_CKSAAP-type-2	0.395	0.539	0.29	0.617	0.33	0.376	0.489	0.239	0.692	0.411
RF_CPCProt	0.417	0.564	0.318	0.536	0.266	0.382	0.502	0.252	0.632	0.378
RF_CTDC	0.392	0.543	0.295	0.57	0.32	0.372	0.497	0.247	0.628	0.39
RF_CTDD	0.384	0.521	0.271	0.625	0.373	0.352	0.461	0.213	0.709	0.475
RF_CTDT	0.406	0.556	0.309	0.544	0.287	0.391	0.515	0.265	0.594	0.346
RF_CTriad	0.41	0.561	0.315	0.542	0.274	0.387	0.505	0.255	0.634	0.371
RF_DDE	0.404	0.545	0.298	0.567	0.313	0.381	0.484	0.234	0.662	0.422
RF_DPC-type-1	0.391	0.533	0.284	0.599	0.345	0.365	0.476	0.227	0.687	0.44
RF_DPC-type-2	0.394	0.537	0.289	0.613	0.333	0.383	0.501	0.251	0.658	0.38
RF_DistancePair	0.387	0.537	0.289	0.585	0.334	0.37	0.491	0.241	0.643	0.405
RF_ESM	0.406	0.541	0.293	0.615	0.324	0.406	0.516	0.266	0.627	0.343
RF_ESM1b	0.404	0.534	0.285	0.631	0.341	0.401	0.514	0.265	0.63	0.347
RF_ESM1v	0.411	0.546	0.298	0.617	0.313	0.411	0.521	0.271	0.635	0.331
RF_ESM2-t12	0.409	0.542	0.294	0.616	0.321	0.409	0.518	0.268	0.625	0.338
RF_ESM2-t30	0.407	0.539	0.291	0.615	0.329	0.411	0.52	0.27	0.612	0.334
RF_ESM2-t33	0.399	0.531	0.282	0.639	0.348	0.396	0.503	0.253	0.653	0.375
RF_ESM2-t36	0.405	0.533	0.285	0.634	0.343	0.402	0.513	0.263	0.637	0.35
RF_ESM2-t6	0.399	0.532	0.283	0.631	0.346	0.401	0.512	0.263	0.625	0.352
RF_FastText	0.415	0.557	0.311	0.568	0.283	0.412	0.521	0.271	0.612	0.331
RF_GDPC-type-2	0.405	0.553	0.306	0.547	0.294	0.383	0.493	0.243	0.645	0.401
RF_GloVe	0.424	0.57	0.325	0.524	0.25	0.416	0.53	0.28	0.582	0.308
RF_PAAC	0.389	0.532	0.283	0.603	0.347	0.381	0.495	0.245	0.642	0.395
RF_PDE	0.364	0.507	0.257	0.649	0.407	0.347	0.456	0.208	0.712	0.487
RF_PRNN	0.417	0.555	0.308	0.59	0.289	0.402	0.507	0.257	0.673	0.365
RF_PTAB	0.412	0.541	0.293	0.642	0.324	0.415	0.522	0.273	0.645	0.327
RF_PTB	0.411	0.541	0.293	0.63	0.324	0.412	0.519	0.27	0.642	0.335
RF_PTBB	0.435	0.575	0.33	0.556	0.238	0.443	0.556	0.309	0.553	0.237
RF_PTU	0.406	0.536	0.287	0.642	0.337	0.405	0.515	0.265	0.638	0.345
RF_PTXLU	0.437	0.568	0.322	0.568	0.256	0.439	0.548	0.3	0.574	0.259
RF_PTXU	0.403	0.533	0.284	0.642	0.344	0.402	0.508	0.258	0.649	0.362
RF_QSOrder	0.39	0.535	0.286	0.6	0.339	0.375	0.487	0.237	0.669	0.416
RF_Seq2Vec	0.401	0.532	0.283	0.649	0.347	0.395	0.503	0.253	0.68	0.376
RF_Word2Vec	0.421	0.565	0.319	0.544	0.264	0.414	0.524	0.274	0.603	0.323
XGB_AAC	0.389	0.547	0.299	0.579	0.309	0.378	0.525	0.275	0.607	0.321
XGB_APAAC	0.381	0.541	0.292	0.587	0.325	0.361	0.491	0.241	0.648	0.405
XGB_ASDC	0.383	0.53	0.281	0.604	0.351	0.351	0.475	0.226	0.674	0.443
XGB_Bepler	0.391	0.529	0.28	0.6	0.354	0.365	0.477	0.228	0.663	0.439
XGB_CKSAAGP-type-1	0.407	0.552	0.305	0.553	0.298	0.38	0.499	0.249	0.621	0.385
XGB_CKSAAGP-type-2	0.421	0.57	0.324	0.517	0.25	0.395	0.509	0.259	0.603	0.362
XGB_CKSAAP-type-1	0.384	0.527	0.277	0.603	0.36	0.365	0.484	0.234	0.651	0.422
XGB_CKSAAP-type-2	0.374	0.521	0.272	0.619	0.372	0.341	0.453	0.205	0.707	0.493
XGB_CPCProt	0.408	0.549	0.301	0.56	0.302	0.359	0.491	0.241	0.637	0.405
XGB_CTDC	0.384	0.539	0.291	0.586	0.327	0.383	0.53	0.281	0.586	0.307
XGB_CTDD	0.372	0.518	0.269	0.62	0.377	0.362	0.482	0.232	0.662	0.427
XGB_CTDT	0.399	0.555	0.308	0.55	0.285	0.377	0.528	0.279	0.587	0.312

XGB_CTriad	0.426	0.575	0.331	0.502	0.237	0.411	0.525	0.276	0.568	0.32
XGB_DDE	0.371	0.518	0.269	0.629	0.378	0.345	0.466	0.217	0.688	0.465
XGB_DPC-type-1	0.389	0.532	0.284	0.595	0.346	0.367	0.486	0.236	0.646	0.417
XGB_DPC-type-2	0.392	0.538	0.289	0.593	0.33	0.364	0.492	0.242	0.642	0.403
XGB_DistancePair	0.389	0.547	0.299	0.579	0.309	0.378	0.525	0.275	0.607	0.321
XGB_ESM	0.367	0.501	0.251	0.648	0.417	0.342	0.454	0.206	0.702	0.492
XGB_ESM1b	0.365	0.499	0.249	0.654	0.424	0.357	0.468	0.219	0.679	0.459
XGB_ESM1v	0.372	0.508	0.258	0.637	0.4	0.363	0.479	0.23	0.66	0.434
XGB_ESM2-t12	0.367	0.494	0.244	0.66	0.433	0.359	0.472	0.223	0.673	0.451
XGB_ESM2-t30	0.367	0.503	0.253	0.645	0.413	0.359	0.472	0.223	0.671	0.449
XGB_ESM2-t33	0.354	0.483	0.233	0.679	0.458	0.342	0.458	0.21	0.7	0.482
XGB_ESM2-t36	0.364	0.5	0.25	0.651	0.421	0.345	0.453	0.205	0.703	0.494
XGB_ESM2-t6	0.352	0.485	0.236	0.674	0.452	0.361	0.482	0.232	0.657	0.428
XGB_FastText	0.37	0.524	0.275	0.616	0.365	0.363	0.482	0.232	0.657	0.428
XGB_GDPC-type-2	0.453	0.6	0.36	0.438	0.168	0.428	0.544	0.295	0.521	0.271
XGB_GloVe	0.395	0.549	0.302	0.57	0.304	0.358	0.474	0.224	0.668	0.446
XGB_PAAC	0.38	0.541	0.293	0.588	0.323	0.36	0.494	0.244	0.64	0.397
XGB_PDE	0.353	0.511	0.261	0.646	0.396	0.341	0.463	0.214	0.693	0.471
XGB_PRNN	0.38	0.516	0.266	0.623	0.382	0.36	0.477	0.227	0.664	0.439
XGB_PTAB	0.364	0.496	0.246	0.655	0.429	0.357	0.465	0.216	0.685	0.467
XGB_PTB	0.37	0.511	0.261	0.634	0.396	0.356	0.474	0.225	0.668	0.445
XGB_PTBB	0.403	0.546	0.298	0.563	0.309	0.41	0.544	0.296	0.536	0.269
XGB_PTU	0.352	0.478	0.229	0.688	0.469	0.372	0.488	0.239	0.648	0.411
XGB_PTXLU	0.415	0.554	0.307	0.544	0.289	0.398	0.52	0.271	0.578	0.333
XGB_PTXU	0.366	0.495	0.245	0.66	0.432	0.365	0.471	0.222	0.675	0.453
XGB_QSOrder	0.393	0.532	0.283	0.597	0.346	0.399	0.512	0.263	0.596	0.352
XGB_Seq2Vec	0.343	0.469	0.22	0.701	0.49	0.327	0.44	0.193	0.724	0.523
XGB_Word2Vec	0.401	0.555	0.308	0.552	0.288	0.362	0.486	0.237	0.648	0.416